ICE 409 • Complete Exam Reviewer

Codes, Standards and Professional Ethics for ICE

Prepared from six uploaded lecture files and two handwritten exam-pointer images. Organized for **remembering, understanding, and applying**. This is a study guide, not a substitute for the adopted standards or an engineering design calculation.

How to use this reviewer

First read the nine topic chapters. Memorize the comparison tables, explain each distinction aloud, then answer the questions without looking at the rationales. Finish with the integrated cases and the quick reviewer. Source citations use physical PDF page numbers, not slide numbers printed inside a page.

Source preservation: Repeated lecture content is consolidated here; explanations and worked examples are added and identified. The companion Source Appendix preserves all five uploaded PDFs page for page, plus a text-and-image extraction of the 48-slide PowerPoint. The original PowerPoint is also embedded as an attachment in that appendix. Source errors remain visible there; this reviewer flags them instead of silently treating them as correct.

Coverage limitation: None of the supplied lectures substantively teaches ASME process-piping standards or ISO quality management. Chapters 1 and 2 therefore provide supplemental background aligned with the handwritten pointers. Their specific exam scope, code editions, and instructor wording cannot be confirmed from the uploads. Other added formulas and engineering explanations are labeled as supplemental where appropriate. External standards and legislation websites could not be accessed from this session; no clause-by-clause external verification is claimed.

The nine-topic coverage map

Topic	Main lecture support	Treatment
ASME standards and process piping	Handwritten pointer only	Supplemental chapter; confirm instructor scope
ISO standards and quality management	Handwritten pointer only	Supplemental chapter; confirm instructor scope
Instrument identification and symbology—ISA 5.1	S1, S2	Lecture-based, with original symbol figures
Process control and instrumentation	S1, S2; S3–S4 for safety comparisons	Basic control concepts explained; added control math labeled
Documentation standards	S1, S2	Lecture-based document selection and revision control
Functional safety—IEC 61508	S3, S4	Lecture-based with numerical clarification
Safety instrumented systems—IEC 61511	S3, S4	Lecture-based with worked SIF/LOPA cases
Hazardous-area classification and protection	S5, S6	Lecture-based; conflicting tables flagged
Electrical and instrumentation installation requirements	S5, S6	Lecture-based; blanket numerical rules qualified

What the handwritten pointers actually say

The first image lists Applying, Understanding, and Remembering. The second lists the nine topics above. Its topic brackets and bottom table appear to associate the first three topics with knowledge items 31–35 and application Part III 51–81; the next four topics with knowledge 36–45, application Part I 1–20, and comprehension Part IV 92–97; and the last two topics with knowledge 46–50, application Part I 21–30, and comprehension Part IV 81–91.

Uncertainty flag: Item 81 overlaps between the handwritten ranges 51–81 and 81–91. The page does not establish a clean total or a complete official blueprint. Practice questions in this reviewer use independent numbering; they are newly written practice, not predictions of actual exam questions.

Source key

ID	Uploaded material	Length and subject
S1	ICE 409 – TOPIC 10.0.pdf	63 pages; ISA symbols and instrumentation documents, despite its filename
S2	ICE 409 – TOPIC 10.1.pptx	48 slides; ISA symbols, tags, documents, revisions
S3	ICE 409 – TOPIC 11.0_compressed.pdf	52 pages; IEC 61508 and IEC 61511
S4	ICE 409 – TOPIC 12.1.pdf	37 pages; functional safety, SIS, and a 20-item quiz
S5	ICE 409 – TOPIC 12.0.pdf	80 pages; hazardous areas and E&I installation
S6	Group 3 – Sharing of Knowledge_compressed.pdf	92 pages; expanded hazardous-area and installation material

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1. ASME Standards and Process Piping

Status: supplemental background. This topic appears in the exam pointers, but substantive ASME material is absent from S1–S6. Learn the distinctions below; do not treat the listed standards as an instructor-confirmed syllabus or assume any particular edition.

1.1 What ASME does

ASME is the American Society of Mechanical Engineers. Its codes and standards address engineering practices, equipment, materials, dimensions, inspection, and safety. A process-piping code supplies rules for an entire piping system; a component standard may specify a flange, fitting, valve, or material-related requirement within that system.

A **code** commonly assembles rules for a defined engineering application. A **standard** commonly establishes consistent technical requirements or practices. The words do not by themselves determine legal force: adoption by a jurisdiction, regulation, contract, or project specification determines what is binding. A voluntary consensus document can become mandatory when adopted.

1.2 Standards you should distinguish

Document	Typical role	Exam distinction
ASME B31.3	Process piping, including chemical, petroleum, pharmaceutical, and related facilities	Process piping rather than every type of piping everywhere
ASME B31.1	Power piping, commonly associated with power-generating and industrial power systems	Distinguish power-piping service from process-piping service
ASME BPVC Section VIII	Pressure-vessel construction rules	A vessel code is not the main process-piping code
ASME B16.5	Pipe flanges and flanged fittings within its stated size/rating scope	Component standard; pressure–temperature rating matters
ASME B16.9	Factory-made wrought butt-welding fittings	Elbows, tees, reducers and similar fittings within scope
ASME B16.34	Valves—flanged, threaded, and welding-end	Valve construction/rating requirements within scope
ASME B1.20.1	General-purpose inch pipe threads	Relevant to applicable NPT threaded connections
ASME BPVC Section IX	Qualification of welding/brazing/fusing procedures and personnel within scope	Qualification does not replace the construction code

Do not confuse: ASME standards specify requirements; ASTM standards often identify material specifications and testing requirements. A correct material designation alone does not demonstrate that the installed piping system meets its governing code.

1.3 The design-to-operation chain

Start by identifying the governing code and service. Establish design pressure, design temperature, fluid hazards, corrosion/erosion allowance, environmental conditions, loading, and operating scenarios. Select compatible pipe, fittings, flanges, valves, gaskets, and bolting. Evaluate supports, flexibility, expansion, vibration, fabrication, joining, examination, pressure testing, and documentation. Exact acceptance criteria depend on the adopted edition and service category.

Design versus operating conditions: Normal operation describes expected routine conditions. Design conditions define the basis used to select and assess the system; they must address the applicable combinations and credible conditions required by the governing code. Do not assume that normal operating pressure is automatically the design pressure, or independently combine every maximum without engineering assessment.

Weakest-link principle: A system is limited by the suitability of its least-capable relevant component at the actual design conditions. A high-rated flange does not make an incompatible gasket or valve acceptable. Corrosion, joint design, temperature derating, and external loads also matter.

1.4 Nominal size, schedule, and pressure class

Term	Meaning	Common trap
NPS / DN	Nominal size designation	Not generally identical to measured inside diameter
Schedule	Wall-thickness series for a given pipe size	Schedule is not itself a universal pressure rating
Pressure class	Rating designation used with material and temperature tables	Class 150 does not mean a universal maximum of 150 psi
Material specification	Required composition, properties, and product requirements	Material compatibility must be assessed for the fluid and temperature
Corrosion allowance	Additional thickness assigned to expected material loss	Not a substitute for chemical compatibility or corrosion management

Increasing schedule usually reduces inside diameter for the same nominal pipe size while outside diameter remains standardized within the applicable dimensional system. Pressure capability still requires the governing design evaluation; wall thickness alone does not establish suitability.

1.5 Fabrication, inspection, and testing

Welding needs qualified procedures and personnel where required. Examination looks for defects using methods appropriate to the code and service: visual examination, radiography, ultrasonic testing, liquid penetrant testing, and magnetic particle testing are examples. The required extent and acceptance criteria vary. A passed pressure test does not replace required examination or demonstrate every aspect of long-term fitness.

Hydrostatic testing uses liquid, typically with much lower stored compressible energy than pneumatic testing. Pneumatic testing uses gas and can release substantial stored energy. Neither method should be improvised from a study guide. **No universal test-pressure multiplier is supplied here:** the uploads do not establish one, and the actual requirement depends on code, temperature, material, system limitations, and test method.

1.6 Why this matters to an ICE student

A transmitter may have the correct range and signal but still be unsuitable if its process connection, diaphragm material, manifold, impulse tubing, seals, or pressure–temperature rating is wrong. Instrument connections are part of the physical pressure boundary. Coordinate the P&ID, instrument data sheet, hook-up, piping specification, and hazardous-area requirements.

Worked application—supplemental: A 4–20 mA pressure transmitter is electrically compatible with the PLC. Its wetted diaphragm is incompatible with the process chemical. Is it acceptable? No. Signal compatibility addresses information transfer; chemical compatibility and pressure-boundary suitability are separate requirements. Review the data sheet and piping specification before selection.

1.7 Practice: remembering, understanding, applying

Q1 — Remembering. Which document is the usual starting point for a chemical plant's process-piping rules? A. ISO 9001; B. ASME B31.3; C. ISA 5.1; D. IEC 61508.

Answer: B. B31.3 addresses process piping. ISO 9001 addresses a quality-management system, ISA 5.1 instrumentation identification/symbols, and IEC 61508 functional safety.

Q2 — Remembering. Does Class 150 mean that every flange is limited to exactly 150 psi at every temperature?

Answer: No. It is a class designation. Applicable pressure–temperature tables, material group, product scope, and service conditions determine the rating.

Q3 — Understanding. Why can a thicker pipe still be unacceptable for a service?

Answer: Thickness does not establish material compatibility, joint integrity, support adequacy, temperature suitability, or compliance with examination and fabrication requirements. Pressure-boundary design is a system assessment.

Q4 — Understanding. Explain why a qualified welder does not by itself prove that a piping installation complies.

Answer: Personnel qualification is one requirement. The construction code also governs design, materials, procedures, fabrication, examination, testing, and records.

Q5 — Applying. A team specifies B31.1 solely because a process line is located inside a power-plant fence. What should you do?

Answer: Verify the actual system, service, code boundary, jurisdiction, and contract requirements. Facility location alone does not establish the correct piping-code scope.

Q6 — Applying. A valve has adequate room-temperature pressure rating but will operate at a much higher temperature. Accept it immediately?

Answer: No. Check its applicable pressure–temperature rating, material and trim compatibility, seals, and design conditions. Room-temperature capability does not establish hot-service suitability.

2. ISO Standards and Quality Management

Status: supplemental background. The handwritten pointers name this topic; the uploaded lectures provide no substantive ISO quality-management lesson. These are general study concepts. Confirm the instructor's standards list and edition before memorizing edition-specific details.

2.1 ISO and the quality-management system

ISO is the International Organization for Standardization. It publishes standards through international consensus. A **quality-management system (QMS)** is the set of policies, processes, responsibilities, resources, and controls an organization uses to meet requirements consistently and improve its performance.

Quality means meeting requirements and intended needs; it is not simply expensive materials or attractive appearance. In instrumentation, quality can mean selecting the specified transmitter, preserving calibration traceability, controlling revisions, recording test results, and correcting recurrent installation errors.

2.2 Standards commonly associated with this topic

Standard	Main purpose	Important distinction
ISO 9000	QMS fundamentals and vocabulary	Explains concepts; not the usual certification requirements document
ISO 9001	QMS requirements	Organizations can seek certification of their QMS against it
ISO 9004	Guidance for sustained organizational success	Broader improvement guidance; distinct from ISO 9001 certification requirements
ISO 19011	Guidelines for auditing management systems	Audit guidance, not a product-certification standard
ISO/IEC 17025	Competence of testing and calibration laboratories	Laboratory accreditation context; not interchangeable with ISO 9001 certification
ISO 14001	Environmental management systems	Environmental management rather than QMS requirements
ISO 45001	Occupational health and safety management systems	Occupational safety management rather than product quality certification

Certification trap: ISO publishes standards; independent certification bodies conduct certification. ISO 9001 certification concerns the organization's QMS within a defined scope. It does not guarantee that every product is defect-free or certify a transmitter's hazardous-area suitability.

2.3 Seven quality-management principles

Principle	Meaning	ICE example
Customer focus	Understand and satisfy applicable requirements	Confirm range, accuracy, delivery, and documentation needs
Leadership	Establish direction and accountability	Assign authority for document approval and release
Engagement of people	Involve competent people	Train technicians and listen to installation feedback
Process approach	Manage related activities as connected processes	Link procurement, calibration, installation, and testing
Improvement	Improve performance continually	Reduce repeated loop-check failures
Evidence-based decision making	Use reliable data	Analyze calibration drift and defect records
Relationship management	Manage relevant interested-party relationships	Evaluate instrument vendors and calibration suppliers

2.4 PDCA and risk-based thinking

Plan: Establish objectives, requirements, resources, responsibilities, risks, and controls. **Do:** Carry out the planned process. **Check:** Monitor, measure, inspect, audit, and compare results against requirements. **Act:** Correct problems and improve the system based on evidence.

For a transmitter installation: plan the approved specification and test procedure; install and calibrate; check loop results and records; investigate failures and improve the process. PDCA is cyclical, not a one-time sequence completed when a certificate is obtained.

Risk-based thinking anticipates what could prevent intended results and what opportunities could improve them. High-consequence failures deserve suitable controls. A formal functional-safety SIL assessment and QMS risk-based thinking are related in purpose but are different activities with different requirements.

2.5 The ISO 9001:2015 clause structure—edition-specific background

Clause	Subject	Typical evidence
4	Context of the organization	Relevant interested parties, scope, processes
5	Leadership	Quality policy, roles, accountability
6	Planning	Risks/opportunities, objectives, planned changes
7	Support	Resources, competence, awareness, communication, documented information
8	Operation	Controlled delivery, design/procurement where applicable, release, nonconforming outputs
9	Performance evaluation	Monitoring, analysis, internal audit, management review
10	Improvement	Nonconformity/corrective action and continual improvement

Clauses 1–3 cover scope, normative references, and terms/definitions. This is a **2015-based study outline**, not a claim that the instructor or project has adopted that edition or that amendments/revisions do not exist.

2.6 QA, QC, correction, and corrective action

Pair	First concept	Second concept
Quality assurance / Quality control	QA provides confidence through planned/systematic processes	QC focuses on fulfilling requirements through operational controls and checks
Correction / Corrective action	Correction fixes a detected nonconformity	Corrective action addresses its cause to prevent recurrence
Verification / Validation	Verification checks specified requirements	Validation checks suitability for the intended use/application
Document / Record	Controlled instructions describe what to do	Retained records provide evidence of what happened; both are documented information
Internal / External audit	Internal audits assess the organization's own system	External audits include customer/supplier and independent certification audits

Example: Recalibrating one out-of-tolerance transmitter is a correction. Investigating why the wrong reference standard was repeatedly used, retraining staff, revising the process, and checking effectiveness are corrective-action activities. Merely writing “operator error” is not adequate cause analysis.

2.7 Measurement traceability and records

Calibration compares an instrument with a reference; adjustment changes its response. They are not synonyms. Metrological traceability links a measurement result to an appropriate reference through a documented unbroken chain of calibrations, each contributing uncertainty. A traceability claim should be supported by records and suitable references; a sticker alone is not enough.

Useful records include instrument tag/serial number, method, reference equipment, relevant traceability, conditions, as-found/as-left results, acceptance criteria, dates, personnel, adjustments, and disposition. Investigate the impact of an out-of-tolerance result on earlier measurements instead of only changing the sticker.

2.8 Practice

Q7 — Remembering. Which standard supplies QMS requirements commonly used for organizational certification? A. ISO 9000; B. ISO 9001; C. ISO 19011; D. ISA 5.1.

Answer: B. ISO 9000 supplies concepts/vocabulary; ISO 19011 supplies auditing guidance; ISA 5.1 addresses instrumentation symbols/identification.

Q8 — Remembering. Expand PDCA.

Answer: Plan, Do, Check, Act. It structures process control and improvement.

Q9 — Understanding. Why does ISO 9001 certification not prove that a specific pressure transmitter is accurate?

Answer: Certification concerns a QMS within its scope. Accuracy requires device-specific specifications, calibration evidence, acceptance criteria, and suitable application conditions.

Q10 — Understanding. A cable label is replaced after an error. Is that automatically corrective action?

Answer: No. Replacement corrects the observed error. Corrective action requires evaluating/addressing the cause and checking effectiveness to prevent recurrence.

Q11 — Applying. Loop-check failures recur because technicians use obsolete drawings. Apply PDCA.

Answer: Plan controlled current drawings and responsibilities; implement distribution controls and training; check audits and failure records; improve the cause of obsolete-copy use and verify effectiveness. Replacing one drawing alone does not close the systemic issue.

Q12 — Applying. A laboratory is accredited to ISO/IEC 17025 for temperature calibration. Can you assume pressure calibration is also accredited?

Answer: No. Accreditation is scope-specific. Check the actual measurement quantities, ranges, methods, and capabilities in its accredited scope.

3. Instrument Identification and Symbology—ISA 5.1

Lecture basis: S1 pp. 2–25; S2 slides 2–25. The lectures reference different ISA 5.1 editions: 1984/reaffirmed 1992, 2009/R2018, and a 2024 product reference. Treat them as mixed-edition teaching material. Use the project's adopted edition and drawing legend for exact conventions.

3.1 Identification versus symbology

Identification uses letters and numbers to state what an instrument/function measures and does. Symbology uses shapes and lines to show function type, location/accessibility, process connections, and signal relationships. Together they form a common engineering language across design, operation, maintenance, and troubleshooting.

An instrument measures and/or controls a variable. A process variable is a property such as flow, pressure, level, or temperature. A loop links instruments/functions through signals to perform measurement or control. A P&ID documents their relationship to the process.

3.2 How a tag is read

TIC-103 = T (temperature) + I (indication) + C (control) + 103 (loop identifier). The first letter identifies the measured or initiating variable. Succeeding letters describe functions and permitted modifiers. Numbers identify the loop, possibly with plant/area information according to the project convention.

Important correction: The lecture graphic presents a “3-letter” pattern for convenience. Tags are not universally three letters, and a fixed second-letter/third-letter rule cannot decode every tag. PT has two letters; TSH has a switch function and high modifier; PDIT contains differential as a variable modifier. Interpret letter position and function using the applicable table and legend.

Letter	As first letter / variable	As succeeding function or modifier
F	Flow	Meaning depends on position and adopted table
T	Temperature	Transmit / transmitter
P	Pressure	Do not substitute an assumed function for the table
L	Level	Low modifier in the appropriate position
A	Analysis	Alarm
S	Speed/frequency	Switch
V	Vibration/mechanical analysis as identified by the source/project	Valve or related final-element function
I	Table-dependent first-letter meaning	Indicate
C	Table-dependent first-letter meaning	Control
R	Table-dependent first-letter meaning	Record

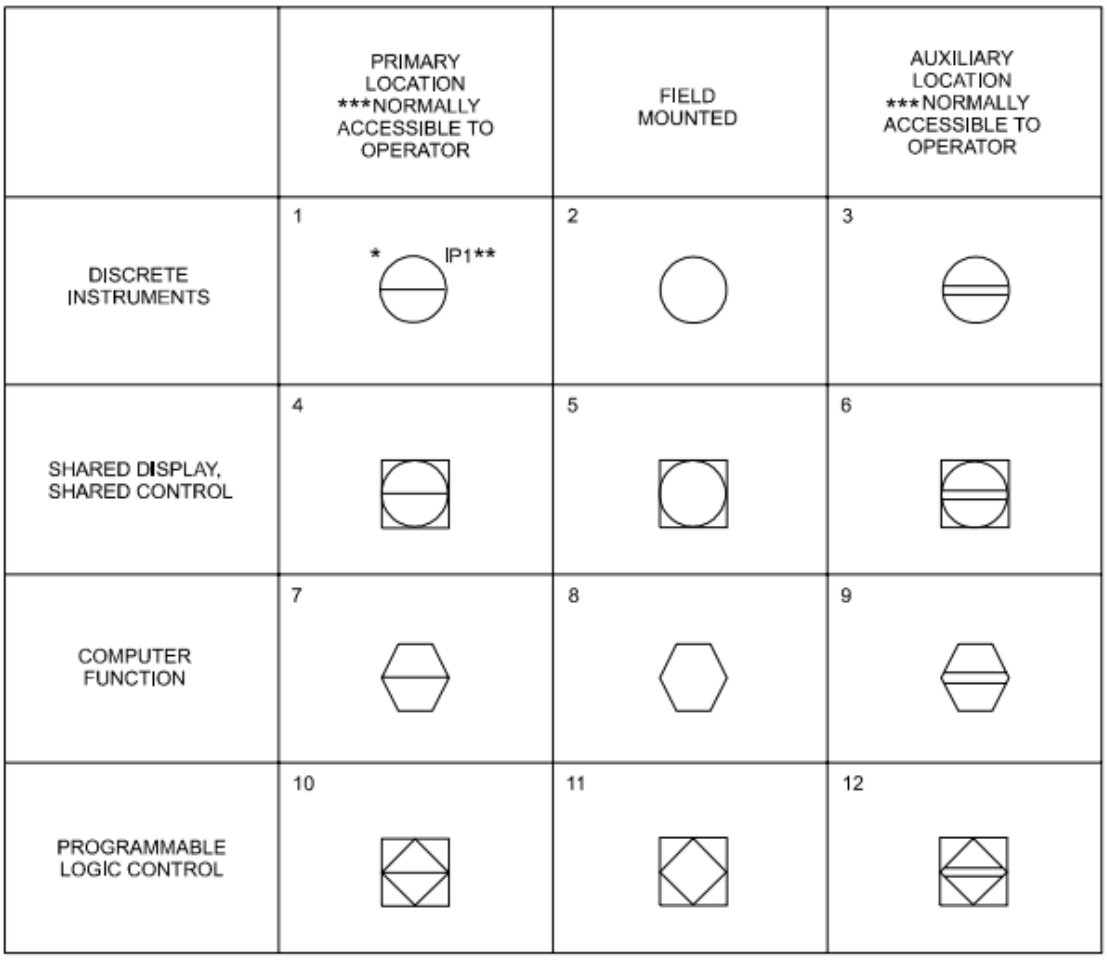
Letter	As first letter / variable	As succeeding function or modifier
D	Table-dependent first-letter meaning	Differential when modifying a measured variable
H	Table-dependent first-letter meaning	High modifier

This is a study subset, not the full ISA identification-letter table. The same letter may have different meanings in different positions; examples are more useful than memorizing one meaning per letter.

Tag	Interpretation	Reasoning
PT-101	Pressure transmitter, loop 101	First P = pressure; T = transmit
FT-102	Flow transmitter, loop 102	Flow is measured and transmitted
LT-103	Level transmitter, loop 103	L is first, so it means level
FIC-101	Flow indicating controller, loop 101	Flow, indication, control
PIT-101	Pressure indicating transmitter	Indicates and transmits pressure
PDT-101	Differential-pressure transmitter	D modifies pressure; T transmits
TSH-101	Temperature switch, high	S = switch; H = high
LAH-101	Level alarm, high	A = alarm; H is a modifier
FV-101	Flow-loop valve/final element	Its first letter follows the loop's initiating variable

Shared loop example: FT-101 → FIC-101 → FV-101. Letters distinguish functions; 101 ties them to the same flow loop. The source mentions reserving 900–999 for safety loops as a possible project convention; **that numbering scheme is not universal ISA law**.

3.3 Instrument bubbles and location



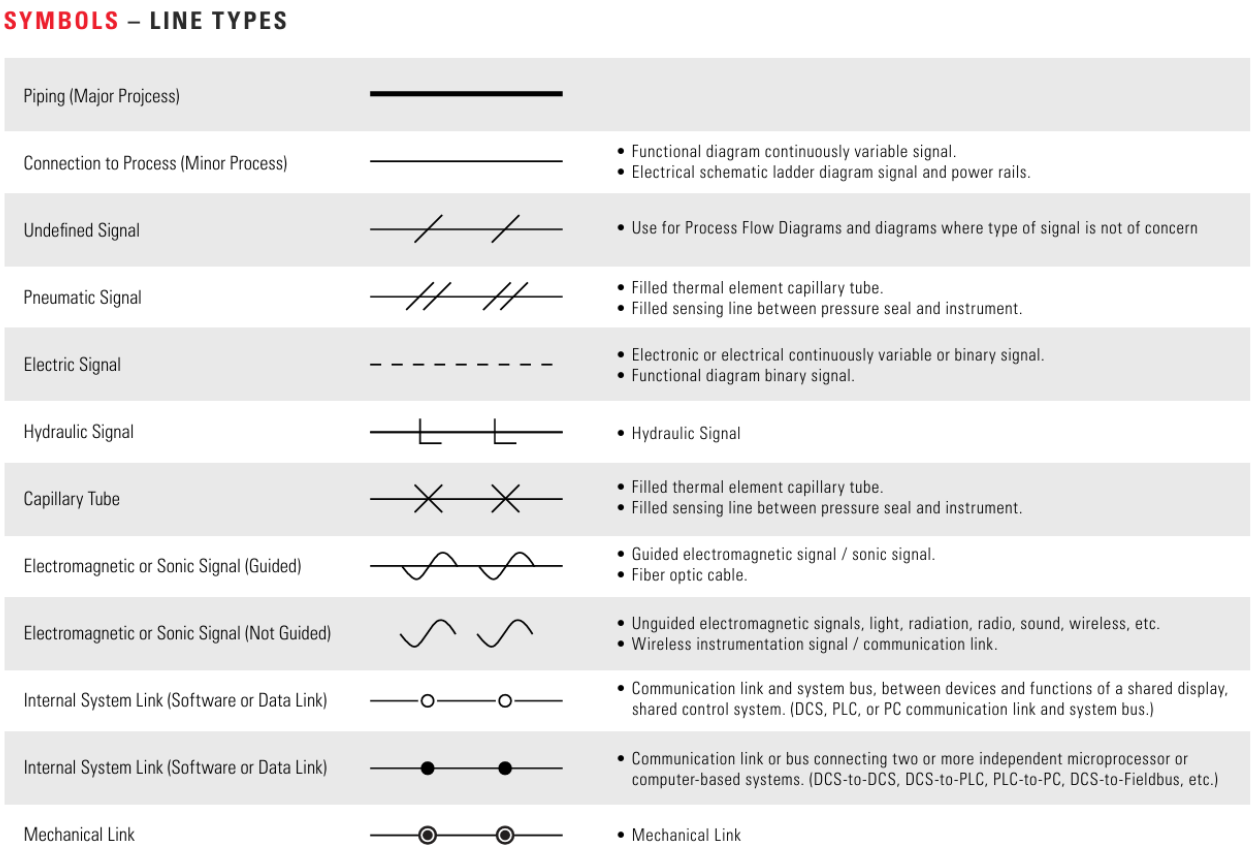
Original lecture figure: instrument/function types and location conventions. S2, media image18. Interpret using the drawing legend and adopted edition.

Common teaching conventions use a plain circle for a discrete field instrument, a circle with a solid horizontal line for an operator-accessible primary location, and a dashed horizontal line for an inaccessible/behind-panel location. Additional lines and enclosing shapes distinguish auxiliary locations, shared display/control, computer functions, and programmable logic functions. **Location/accessibility is not the same thing as process variable.** The letters identify the variable/function; the graphic supplies other information.

The source also includes an older location table using “panel mounted,” “field mounted,” “auxiliary room,” and “behind panel” wording. These figures are retained in the Source Appendix. Do not combine symbols from different editions without checking their legend.

3.4 Process and signal lines

Line category	What it represents	Exam interpretation
Process piping / connection	Physical process-fluid path or connection	Follow material through equipment, piping, and valves
Electrical/electronic signal	Information carried electrically	Follow transmitter/controller I/O connections
Pneumatic signal	Information conveyed through air pressure	Distinguish it from a process-fluid pipe
Hydraulic signal	Information conveyed through liquid pressure	Identify the signal medium in the legend
Capillary connection	Filled capillary system	Do not assume it is an ordinary electrical signal
Data/software link	Communications or internal system connection	Not a physical pipe carrying process fluid
Mechanical link	Mechanical transmission/connection	Linkage rather than electrical wiring
Undefined signal	Signal type not specified	Do not invent a medium



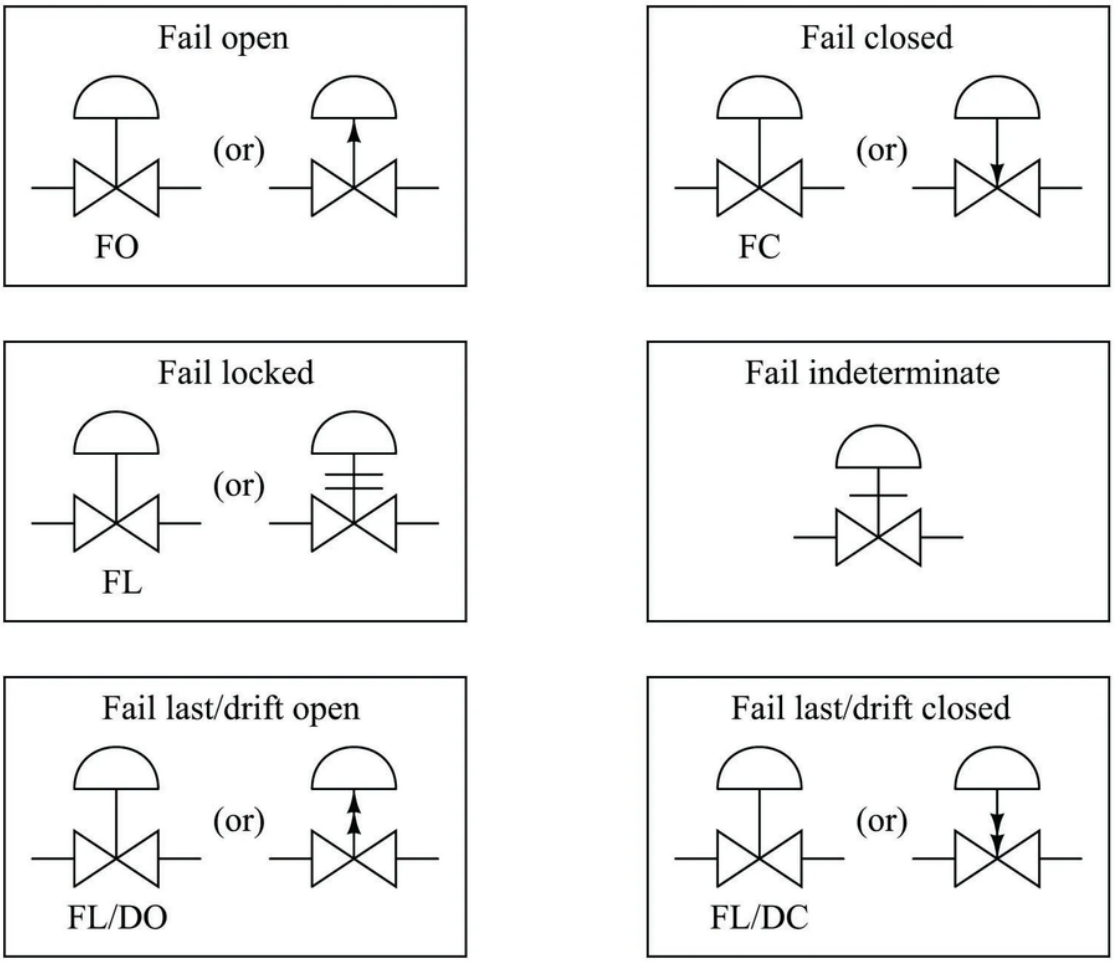
Original lecture line-symbol chart, S2 media image16. Line styles are reproduced for study; the applicable drawing legend controls.

Mnemonic: Process line = material; signal line = information. Pneumatic and hydraulic signals use physical media, but their diagrammatic purpose is transmission of a measurement or command.

3.5 Valves, actuators, and failure action

Valves direct, isolate, regulate, or prevent reverse flow. Common examples include gate, globe, ball, butterfly, check, relief, and safety valves. A check valve is intended to oppose reverse flow; a globe-style control valve is commonly used for throttling. Actual service suitability depends on design and application.

An actuator receives a command, uses an energy source, and produces mechanical motion. Pneumatic actuators use compressed air; hydraulic actuators use pressurized liquid; electric actuators use electrical power. Motion can be linear or rotary. A valve symbol and its actuator symbol convey different information.



Original valve and actuator failure-action symbols, S2 media image15.

Failure action	Meaning	Typical teaching application / caution
FO—fail open	Moves to open on the specified failure	Cooling flow may need to continue; verify hazard analysis
FC—fail closed	Moves to closed on the specified failure	Fuel feed may need to stop
FL—fail locked	Holds/locks its position on the specified failure	Requires an actual locking/holding arrangement
Fail indeterminate	Final position is not assured	Do not claim a guaranteed safe position
FL/DO	Initially holds last position, then drifts open	Temporary hold is not permanent lock
FL/DC	Initially holds last position, then drifts closed	Account for loss of stored energy/leakage

Qualification to the lecture: FO and FC describe the specified failure behavior; they do not prove every actuator uses a spring. Fail-safe behavior depends on the complete actuator/accessory/energy arrangement, process forces, and which failure is being considered. “Open is always safer” and “closed is always safer” are both wrong.

3.6 Read a P&ID systematically

Identify equipment and the process flow direction. Locate the measured variable and instrument tag. Decode the letters and loop number. Read bubble/location and line styles using the legend. Trace the signal to the controller, then to the final element, and back to the process. Check valve failure action, alarms, trips, and interlocks. Finally confirm that related documents use the same tags and revision.

Worked application: FT-101 measures flow; FIC-101 compares it with a setpoint; FV-101 changes flow. If FIC-101 is showing a bad measurement, a P&ID helps identify the loop, but a loop diagram is needed to trace exact terminals and power connections. The P&ID does not establish every wiring detail or physical distance.

3.7 Practice

Q13 — Remembering. Decode TIC-103.

Answer: Temperature indicating controller, loop 103. T is the initiating variable; I and C are functions; the number identifies the loop.

Q14 — Remembering. What does T mean in PT-101? A. Temperature; B. Transmitter; C. Timer; D. Trip setting.

Answer: B. Letter position matters: P identifies pressure; succeeding T denotes transmission.

Q15 — Understanding. Why is “every instrument tag contains exactly three letters” incorrect?

Answer: The source's three-letter diagram is a simplified example. Tags may have different lengths and modifiers; PT, TIC, PDT, and TSH require different parsing.

Q16 — Understanding. Why does a line connecting two symbols not automatically mean fluid flows between them?

Answer: It may be an electrical, pneumatic, hydraulic, mechanical, or data signal. Its style and the legend establish its meaning.

Q17 — Applying. A cooling-water valve must maintain cooling after loss of air. Which failure action should be considered?

Answer: FO, provided the hazard analysis and complete valve/actuator design establish that opening is the required safe response. The service name alone does not certify the final choice.

Q18 — Applying. PT-101 and FIC-101 appear in what is labeled a simple flow-control loop. Is matching 101 sufficient to prove they are correct?

Answer: No. Check the intended variable and project convention. PT measures pressure while FIC controls flow; differential-pressure flow measurement can justify particular tagging conventions, but the shared number alone does not prove consistency.

4. Process Control and Instrumentation

Lecture basis: S1 pp. 4–8, 17–25; S2 slides 3, 17–25 and 37; S3 pp. 24–26 and 44 for BPCS/SIS. Feedback terminology, signal-scaling math, and detailed controller explanations below are supplemental teaching explanations rather than a full control-theory lecture supplied by the files.

4.1 The purpose of a control loop

A control loop measures a process variable, compares it with the desired value, and manipulates something to reduce an unwanted deviation. A sensor detects the physical quantity; a transmitter converts/sends a usable signal; a controller makes a decision; a final control element changes the process. Instruments may combine functions, so four named functions do not always mean four separate physical devices.

Term	Meaning	Tank-level example
PV—process variable	Measured quantity	Actual liquid level
SP—setpoint	Desired target	Target level
Error	Difference between target and measurement under the defined sign convention	SP − PV in a common convention
MV—manipulated variable	Quantity intentionally changed	Inlet flow, changed by the inlet valve
Disturbance	External effect moving the process away from target	Outlet-flow change
Feedback	Uses measured process response	Measure actual level before adjusting inlet
Open loop	Does not correct based on measured controlled-variable feedback	Fixed valve command with no level correction

Closed-loop sequence: Process → measurement → controller → final control element → process. The source documents this relationship; detailed design also needs dynamics, action, range, failure behavior, and limits.

4.2 Manual, automatic, and control action

In manual mode, the operator directly adjusts the controller output or final-element command. In automatic mode, the controller adjusts output based on the configured control algorithm and measured input. A controller in manual may still display the PV; showing a value does not prove automatic control is active.

Direct/reverse controller action describes how controller output responds to an increase in measurement for a fixed setpoint under the stated convention. Choose action together with valve/process behavior to create **negative feedback**. A rising level with an inlet-control valve normally needs reduced inlet flow; whether controller output must rise or fall depends on whether increased output opens or closes that valve.

Supplemental PID summary: Proportional action responds to present error; integral action accumulates error and can remove steady offset; derivative action responds to the rate of change and can be sensitive to noise. Do not assume that every loop needs all three or that higher gain always improves control. The uploads do not provide tuning rules or required numerical tuning methods.

4.3 Signals and I/O

Signal / I/O	Typical meaning	Example
AI—analog input	Numeric measurement received by controller	Pressure transmitter, 4–20 mA
AO—analog output	Numeric command sent by controller	Valve position/current command
DI—digital/discrete input	State received	Limit switch, run contact
DO—digital/discrete output	State command sent	Start relay, solenoid command
RTD / thermocouple input	Sensor-specific temperature measurement	Requires appropriate input type and wiring
Communications	Digitally encoded data	Fieldbus/network measurement or command

The lectures name 4–20 mA, digital signals, RTD/thermocouple signals, and communication signals. A “digital signal” can mean a discrete state or encoded communications; context matters. A PLC or DCS may handle routine control, but ordinary controller hardware is not automatically an acceptable safety logic solver.

4.4 Worked 4–20 mA scaling—supplemental

For a linear transmitter with lower range value LRV and upper range value URV:

Fraction of span = (I − 4 mA) / 16 mA. PV = LRV + [(I − 4) / 16] × (URV − LRV). I = 4 + 16 × (PV − LRV) / (URV − LRV), in mA.

For a 0–100 °C range at 12 mA, fraction = (12 − 4)/16 = 0.5; temperature = 50 °C. For a −50 to 150 °C range at 8 mA, fraction = 0.25; temperature = −50 + 0.25 × 200 = 0 °C. Subtract the lower range before scaling; span and upper-range value are not interchangeable.

The live zero of 4 mA distinguishes a valid lower-range measurement from some loss-of-signal conditions. However, zero current can result from several faults, and fault thresholds depend on device configuration and applicable conventions. **4 mA does not necessarily mean zero physical units**; it means the configured LRV. Nonlinear/square-root applications require the configured transfer function rather than blindly applying this linear formula.

4.5 Alarm, permissive, interlock, and trip

Function	Purpose	Example
Alarm	Notify an operator of an abnormal condition requiring response	High tank level alarm
Permissive	Required condition allowing an operation to start/proceed	Start allowed only if lubrication is available
Interlock	Enforce a dependency or protective constraint	Prevent incompatible valves from being open together
Trip	Initiate a protective action when defined conditions occur	High-high pressure closes feed valve

A permissive commonly prevents starting; a running interlock/trip can stop an already-operating system. Actual meanings must be specified in the control narrative. Not every alarm is an independent protection layer, and not every interlock is a safety-rated SIF.

4.6 Logic and simple truth tables

AND requires all required conditions; OR needs at least one; NOT inverts a condition. The lecture pump-start example requires both conditions to be satisfied. Timing, reset, latching, bypasses, and fault states must also be defined in a real implementation.

A	B	A AND B	A OR B
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	1

Application: Start permission = suction available AND lubrication healthy. If either is false, permission is false. A start command is a separate input; permission alone need not start the motor. After a trip, distinguish clearing the fault, resetting a latch, and issuing a new start command.

4.7 Troubleshooting a loop

First confirm the operating mode and the real process condition. Compare the local indication with the control-system reading. Check range, units, signal type, power, polarity, wiring, input configuration, and device status. Trace the output to the final element and verify response/feedback. Use approved isolation and test procedures; injecting a signal into an operating safety loop can cause a real trip or hide a real hazard.

Separate an accurate measurement with poor control tuning from an incorrect measurement, a wiring fault, or a stuck valve. Calibration, loop checking, functional testing, and tuning address different problems.

4.8 Practice

Q19 — Remembering. Which variable is intentionally adjusted to influence the PV?

Answer: The manipulated variable. For example, inlet flow is manipulated to influence vessel level.

Q20 — Remembering. A pressure transmitter connected to a PLC normally uses which I/O category? A. AI; B. AO; C. DO; D. Motor feeder.

Answer: A. The controller receives a numeric measurement. An AO sends a numeric command; a DO sends a discrete command.

Q21 — Understanding. Why can the operator see the correct PV while automatic control is not functioning?

Answer: Measurement/display and automatic output calculation are different functions. The loop may be in manual, the output may be limited, or the final element may not respond.

Q22 — Understanding. Is an alarm identical to a trip?

Answer: No. An alarm notifies; a trip initiates a defined protective action. Some systems combine them, but annunciation alone does not establish an automatic shutdown function.

Q23 — Applying. A linear 0–200 kPa transmitter sends 16 mA. What is the indicated pressure?

Answer: (16 − 4)/16 = 0.75 span; 0.75 × 200 = **150 kPa**. Using 16/20 would produce an incorrect 160 kPa result.

Q24 — Applying. A high tank level rises further because its inlet valve opens more after the controller responds. What is the likely control issue?

Answer: The complete loop may have the wrong action and be producing positive feedback. Verify controller action, output scaling, actuator/positioner behavior, and process response before changing settings.

5. Documentation Standards

Lecture basis: S1 pp. 26–60; S2 slides 26–46. ISA 5.1 supports symbols/identification. The lecture references ISA 5.2 for binary logic. ISA 5.4 (loop diagrams) and ISA 20 (instrument specification forms) are useful supplemental references; their editions and exact project applicability are not established by the uploads.

5.1 One plant, multiple document purposes

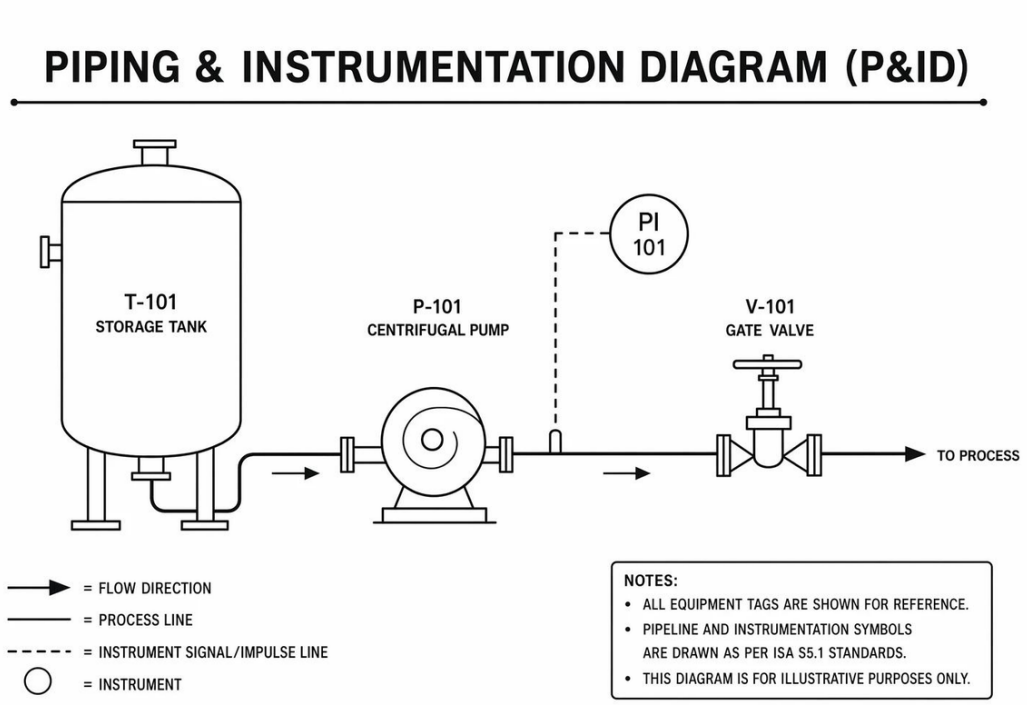
No single drawing answers every question. Select the document by the information needed, then cross-check the tag and revision across related documents. A correct P&ID with an obsolete loop drawing can still lead to an installation error.

Document	What it primarily answers	Typical contents
Process Flow Diagram—PFD	What are the main process steps and streams?	Major equipment, major streams, key operating conditions
Piping and Instrumentation Diagram—P&ID	How is the process connected, monitored, and controlled?	Piping, valves, equipment, instruments, control relationships
Instrument index	What instruments exist in the project?	Tags, services, types, locations, drawing references
Instrument data sheet	What must this device technically satisfy?	Range, accuracy, process conditions, wetted materials, connections, signals
Instrument loop diagram	How is this loop electrically/signal-connected?	Field device, wires, terminals, JB, power, PLC/DCS I/O, final element
Hook-up / installation detail	How is it physically mounted and connected?	Tubing, fittings, manifolds, supports, process connections
I/O list	Which controller channels handle which signals?	Tags, signal types, I/O type, channel/address information
Wiring diagram	How are electrical devices interconnected?	Equipment, conductors, power and signal connections
Terminal/connection diagram	Where does each wire terminate?	Terminal designations, strip numbers, wire/core references
Cable schedule	Which cable runs from where to where?	Cable ID, source, destination, type, size, cores, sometimes length
Control narrative / philosophy	How should the system behave?	Modes, sequences, actions, alarms, interlocks, abnormal responses
Logic / functional diagram	How are conditions combined to produce actions?	Gates, logic functions, timing, latching or control relationships
Cause-and-effect diagram	Which causes produce which outputs/actions?	Input events versus shutdown, alarm, or other responses
Alarm and setpoint list	What thresholds and settings apply?	Alarm limits, trip points, control setpoints, units, tags

Document	What it primarily answers	Typical contents
Location/layout drawing	Where is equipment physically installed?	Instrument/panel/JB locations and mounting layout
Calibration and test records	What was checked and what were the results?	Dates, results, adjustments, acceptance, maintenance history

5.2 PFD versus P&ID versus loop diagram

A PFD describes the broad process. A P&ID adds functional detail about piping, valves, equipment, and instrumentation. A loop diagram follows one measurement/control circuit to the terminal and I/O level. A P&ID is not necessarily to scale and does not ordinarily specify every cable core, terminal, route length, or construction dimension.



Original illustrative P&ID from S2, media image23. It shows tank, pump, pressure indication, valve, and flow direction; its own note says it is simplified.

Document-choice example: “Where is PT-101?” may require the location drawing. “What range does it need?” requires its data sheet. “Which PLC terminal receives it?” requires the loop/termination/I/O documentation. “What happens at high-high pressure?” requires the narrative, cause-and-effect, and relevant safety requirements.

5.3 Index and data-sheet consistency

The instrument index is a project-wide master list, not a complete specification for every device. Some projects include setpoints, IP ratings, I/O types, or hazardous-area data in the index; those fields are project-specific. The data sheet supplies detailed technical requirements for one device. Supplier documents must be checked against the approved requirements, not simply accepted because a tag matches.

For a pressure transmitter, check tag/service, measured range and units, operating/design conditions, accuracy, process fluid, wetted materials, pressure connection, electrical signal, power supply, hazardous-area certification, enclosure rating, and drawing references. Do not confuse measuring span with

overpressure capability.

5.4 Logic diagrams, narratives, and cause-and-effect

A logic diagram is a graphic description of input/condition/output relationships; it is useful for PLC, relay, interlock, and alarm logic. A control narrative explains behavior in readable words. A cause-and-effect matrix maps defined initiating conditions to required actions. They should agree about setpoints, delays, modes, priorities, reset conditions, and failure responses.

Example—pump permissive: The narrative says that the pump may start only with sufficient suction level and lubrication pressure. The logic uses AND between those conditions. A separate start command and any latch/reset requirements must be specified. If a running trip is intended on loss of lubrication, document that action explicitly; a start permissive does not automatically prove a running trip exists.

5.5 Revision control

The lecture workflow is: **change identified → engineering review → document updated → revision number assigned → checked/approved → new revision issued → previous revision archived**. Update related documents as a coordinated set, preserve traceability, and ensure users can identify the current approved issue.

The lecture example adds PT-18: determine its location/specification; add it to the P&ID and related documents; move P&ID Revision 3 to Revision 4; check and approve; issue Revision 4; archive/mark Revision 3. A redline can record a proposed/as-built change but is not automatically an approved final drawing.

Without revision control	With effective revision control
Obsolete P&IDs may be used	Current issue is identifiable
Changes are hard to trace	Changes, review, and approval are recorded
Maintenance may use incorrect information	Technicians receive the applicable approved information
Documents contradict one another	Related documents are checked for consistency

5.6 A practical consistency checklist

Match tag, service, loop number, variable, signal, range/units, alarm/trip settings, valve action, power, I/O address, terminals, cable number, hazardous-area suitability, and revision status. Update the instrument index, data sheet, P&ID, loop drawing, I/O list, cable/termination records, narrative/logic, cause-and-effect, and test records as affected. The scope depends on the actual change; do not mechanically revise unaffected documents.

5.7 Practice

Q25 — Remembering. Which document is the project-wide master list of instruments?

Answer: The instrument index. A data sheet describes one device in detail; the index inventories the project.

Q26 — Remembering. Which document best shows tubing, manifolds, mounting, and process connection details?

Answer: The hook-up/installation drawing. A loop diagram primarily addresses circuit/signal interconnections.

Q27 — Understanding. Why is a P&ID insufficient for tracing a wire to a specific terminal?

Answer: Its primary role is process/control relationships. Loop, wiring, terminal, and I/O documents provide the detailed connection information.

Q28 — Understanding. Why should the old revision be archived rather than left as an indistinguishable working copy?

Answer: Archiving preserves traceability while preventing obsolete information from being mistaken for the current approved issue.

Q29 — Applying. A transmitter range changes from 0–10 bar to 0–16 bar. Name the affected checks.

Answer: Review the data sheet, calibration, controller scaling, displays, alarms/trips and safety requirements where affected, index fields, test procedures, and relevant drawings. A correct physical installation with old scaling can still report the wrong value.

Q30 — Applying. The P&ID says FC but the narrative says the valve must remain open on loss of air. Which should you silently follow?

Answer: Neither should be silently preferred. Raise the conflict for engineering resolution against the required safe state; correct the controlled documents and verify the actual equipment behavior.

6. Functional Safety—IEC 61508

Lecture basis: S3 pp. 2–21, 45–47; S4 pp. 2–20. The lectures present IEC 61508 as the general E/E/PE framework and IEC 61511 as its process-sector application. S3 cites the 2010 second edition; the uploads do not establish a universally adopted project edition.

6.1 What functional safety means

Functional safety is the part of overall safety that depends on a system responding correctly to its inputs and performing the required safety function. The lecture contrasts passive insulation on a hot motor with an overtemperature sensor/trip. The insulation is passive protection; the sensing-and-trip response is a functional-safety measure.

IEC 61508 addresses electrical, electronic, and programmable electronic safety-related systems. It is a multi-sector foundation for sector-specific standards. The purpose is to reduce risk to a tolerable level through lifecycle management of random hardware failures and systematic failures—not to promise zero risk.

Core sequence: Hazardous condition → detection → decision → protective action → safe state. Safety depends on the whole chain. A reliable safety PLC cannot compensate for a sensor that never detects the hazard or a valve that never moves.

6.2 Essential vocabulary

Term	Meaning	Distinction
Hazard	Potential source of harm	A hazard exists even before an accident happens
Risk	Combination of likelihood and consequence in the relevant assessment	Hazard severity alone does not determine risk
Safety function	Function intended to achieve/maintain a safe state for a defined hazard	Must state what action is required and when
Safety integrity	Confidence/performance that the safety function will be achieved as required	Not merely the function's description
SIL	Safety Integrity Level	Discrete integrity level associated with a safety function
Safe state	Defined condition in which the assessed hazard is adequately controlled	Not always “everything off”
E/E/PE	Electrical/electronic/programmable electronic	Technology scope of IEC 61508
ALARP	As Low As Reasonably Practicable	Risk-reduction decision principle; legal/application context matters

Source terminology warning: S3 p. 28 says “Safety Integrated Level”; the correct expansion is **Safety Integrity Level**. S4 uses SRS to mean safety-related system, while process-safety work often uses SRS for **Safety Requirements Specification**. Define the acronym from context instead of assuming only one meaning.

6.3 Risk assessment and reduction

Identify hazards and hazardous events; analyze their likelihood and consequences; evaluate whether risk is tolerable; determine additional risk reduction; implement and verify/validate the measures. Consider inherently safer process design, BPCS, alarms with operator response, SIS, and other protection. A listed safeguard earns numerical risk-reduction credit only when its relevant effectiveness, independence, integrity, and assumptions are justified.

Method	Main question	Character
HAZOP	What deviations could cause hazards or operability problems?	Structured guide-word analysis of nodes and parameters
Risk graph	What target integrity follows the calibrated decision path?	Generally qualitative/semi-quantitative, method-dependent
LOPA	Is the residual event frequency acceptable after independent layers?	Semi-quantitative scenario analysis
SIL verification	Can the proposed SIF design meet the assigned integrity requirements?	Design/performance assessment, not initial hazard identification

The lectures describe risk graphs using consequence, exposure/frequency, avoidance, and occurrence parameters. Their exact definitions and calibration vary. A generic risk graph is not a universal scoring chart. LOPA results are estimates dependent on scenario and assumptions, **not exact physical predictions**.

6.4 SIL: low-demand metrics

For low-demand functions, **PFDavg** is the average probability of dangerous failure on demand. Under the usual low-demand interpretation, **RRF ≈ 1/PFDavg**. The table below gives the standard study ranges, added as numerical clarification to the lecture's simplified RRF chart.

SIL	PFDavg range, low demand	Approximate corresponding RRF
1	$10^{-2} \le \text{PFDavg} < 10^{-1}$	>10 to ≤100
2	$10^{-3} \le \text{PFDavg} < 10^{-2}$	>100 to ≤1,000
3	$10^{-4} \le \text{PFDavg} < 10^{-3}$	>1,000 to ≤10,000
4	$10^{-5} \le \text{PFDavg} < 10^{-4}$	>10,000 to ≤100,000

The lectures round these bands to “10–100,” “100–1,000,” and so on. Use the PFDavg inequalities when boundaries matter. S4 p. 6 prints “100,00” at the SIL 4 upper RRF boundary; this appears to be a typographical error for 100,000.

Worked example: PFDavg = 0.004 = 4×10^{-3}; RRF ≈ 250. This lies within the SIL 2 low-demand numerical band. It does **not alone prove SIL 2 compliance**: systematic capability, architecture, common-cause failures, independence, lifecycle controls, and other applicable requirements still matter.

Boundary example: PFDavg = 0.01 lies in SIL 1, not SIL 2, because SIL 2 requires a value strictly less than 0.01. A requirement for a certain reduction must be evaluated with the actual acceptance criterion and engineering margin; avoid ambiguous “at least 100 means SIL 2” shortcuts.

6.5 High-demand/continuous mode

High-demand or continuous functions use **PFH**, the average frequency of dangerous failure per hour. It is not numerically interchangeable with PFDavg. The same SIL number corresponds to a different metric table depending on operating mode.

SIL	PFH band, h^{-1}—supplemental numerical reference
1	$10^{-6} \leq PFH < 10^{-5}$
2	$10^{-7} \leq PFH < 10^{-6}$
3	$10^{-8} \leq PFH < 10^{-7}$
4	$10^{-9} \leq PFH < 10^{-8}$

IEC 61508 defines SIL 1–4. IEC 61511 process-sector practice generally addresses SIFs through SIL 3; a need for reduction beyond the covered level demands additional risk-reduction consideration. Do not infer that SIL 4 is an ordinary process-SIS design target simply because IEC 61508 has four levels.

6.6 The safety lifecycle

The lecture's condensed lifecycle is: concept/boundaries → hazard and risk analysis → safety requirements/risk reduction → design/implementation → installation/commissioning → validation, operation and maintenance → modification/decommissioning. These summaries are educational groupings, not a verbatim reproduction of every formal lifecycle box.

Safety management, competence, planning, verification, assessment, documentation, and configuration control extend across the lifecycle. A change to a trip setpoint, proof-test interval, sensor type, software, bypass practice, or final element may require reassessment. Safety is not complete when equipment is delivered.

6.7 Hardware considerations—IEC 61508-2

Consideration	Study explanation	Important limitation
Random hardware failures	Physical component failures occurring during service	Need credible failure data and applicable models
Safe Failure Fraction—SFF	Proportion involving safe failures and dangerous detected failures in the relevant classification	High SFF alone does not prove a required SIL
Hardware Fault Tolerance—HFT	Ability to tolerate a specified number of hardware faults while retaining the function	Architecture, fault definitions, and method matter
Redundancy	More than one channel/component supports a function	Shared causes can defeat all channels
Diagnostic coverage	Fraction of relevant dangerous failures detected by diagnostics	Detection must lead to appropriate action and timely repair
FMEA / FMEDA	Analyze failure modes/effects; FMEDA includes diagnostic analysis	Not a substitute for realistic application assumptions
Type A / Type B	Simplified distinction between sufficiently understood and more complex failure behavior	Formal classification criteria go beyond “relay versus electronics”

Supplemental relationship: SFF is commonly expressed as (safe failure rate + dangerous detected failure rate) / (safe + dangerous detected + dangerous undetected failure rates), under the applicable definitions. Do not apply the formula without consistent failure classifications and the selected assessment route.

6.8 Software considerations—IEC 61508-3

Specify software safety requirements; design architecture and modules; implement controlled code/configuration; verify modules and integration; validate the software contribution; control operation, maintenance, and modification. Maintain traceability from requirement to implementation to test. Tools, languages, configuration, and competency matter.

Software errors are primarily systematic: they arise from specification, design, coding, configuration, or process errors rather than electronic parts wearing out. Redundant processors running the same defective logic can fail together. IEC 62443 is named in S3 as industrial cybersecurity background; cybersecurity supports safety but is a separate requirements framework.

6.9 Verification, validation, and maintenance

Verification asks whether each phase output meets its specified requirements. Validation asks whether the implemented safety functions meet the safety requirements for the intended application. SIL verification assesses whether the proposed SIF can meet its target integrity; it is not the same as pressing a test button once.

The lectures list risk-based, preventive, condition-based, predictive, and corrective maintenance. Maintain proof-test schedules, repair time assumptions, diagnostic response, competence, bypass control, and records. Repeatedly extending a test interval without recalculation can undermine the claimed integrity.

6.10 Practice

Q31 — Remembering. What does IEC 61508 principally address?

Answer: Functional safety of E/E/PE safety-related systems. IEC 61511 is its process-sector application; ISA 5.1 concerns symbols and identification.

Q32 — Remembering. Which metric is normally used for a low-demand SIF? A. PFDavg; B. Pipe schedule; C. IP rating; D. Cable ampacity.

Answer: A. PFDavg measures average dangerous failure probability on demand. PFH serves high-demand/continuous mode.

Q33 — Understanding. Does “higher SIL” mean an event has a more severe consequence in every case?

Answer: No. SIL is an integrity/risk-reduction requirement for a safety function. Target determination also depends on likelihood, exposure, other layers, and tolerable risk—not severity alone.

Q34 — Understanding. Why can identical redundant controllers fail together?

Answer: A common specification/software error, shared supply, environmental cause, or other common cause can affect both. Redundancy does not eliminate systematic or common-cause failures.

Q35 — Applying. A low-demand SIF has PFDavg = 0.002. Find its RRF and numerical SIL band.

Answer: RRF ≈ 1/0.002 = **500**; numerical band **SIL 2**. Full compliance still requires the applicable architecture, systematic, lifecycle, and application requirements.

Q36 — Applying. A team proves its safety function works once and declares SIL 3. Is the conclusion supported?

Answer: No. One functional success does not establish dangerous failure probability, systematic integrity, architecture, response under faults, maintenance assumptions, or lifecycle compliance.

7. Safety Instrumented Systems—IEC 61511

Lecture basis: S3 pp. 22–52; S4 pp. 21–31. IEC 61511 applies functional-safety principles to SIS in the process-industry sector, including design, installation, validation, operation, maintenance, modification, and decommissioning.

7.1 SIS, SIF, and SIL

Term	Meaning	Vessel-overpressure example
SIS	Safety Instrumented System: the protective system implementing one or more SIFs	Sensors, safety logic solver, final elements, and supporting arrangements
SIF	Safety Instrumented Function: one defined protective function	On high-high pressure, close feed in the required response time
SIL	Safety Integrity Level: integrity requirement for that function	Target assigned from risk assessment and verified for that function
BPCS	Basic Process Control System: routine control	Regulate pressure/flow during production

The safe state is hazard-specific. Closing fuel feed may be right for one scenario; preserving cooling or opening a depressurization path may be right for another. A single SIS can implement multiple SIFs with different integrity requirements. A component's SIL capability/certificate does not make every assembled SIF achieve that SIL.

7.2 The architecture

Process → sensor(s) → logic solver → final element(s) → safe state. The sensor detects the hazardous variable; the logic solver evaluates defined logic; the final element performs the protective action. Consider power, instrument air, wiring, solenoids, process connections, and communications where failure affects the function.

Element	Lecture examples	Strengths and limitations
Sensor/initiator	Pressure/temperature transmitters, level switches, flame/gas detectors	Fast measurement; may drift, foul, plug, or fail in harsh conditions
Logic solver	Safety PLC, safety controller, certified logic or suitable relay system	Diagnostics and programmable logic; configuration, software, shared causes and cybersecurity matter
Final element	Shutdown valve, motor contactor, damper, circuit breaker, blowdown valve	Physical protective action; may stick, leak, move too slowly, or lose energy supply

The lecture calls final elements the “weakest link/highest PFD.” They often contribute substantially, but **this is not a universal mathematical law**. Actual subsystem contributions depend on data, architecture, test coverage, conditions, and maintenance.

7.3 BPCS versus SIS and independence

The BPCS controls routine operation; the SIS supplies a defined risk-reduction function. Independence is required to avoid the same failure defeating both normal control and credited protection. Assess shared sensors, process taps, logic, I/O, communications, power, utilities, software, maintenance practices, and environment.

Qualification: S3 states “strict physical and functional independence.” Treat that as a teaching simplification, not a blanket rule that every item must always be on a separate physical platform regardless of design. The actual permitted sharing and required measures depend on the adopted standard and justified independence/common-cause assessment. Do not credit two layers as independent solely because they have different names.

7.4 Specify the SIF before selecting hardware

A safety requirements specification should define the hazardous event, initiating conditions, trip thresholds, action, safe state, response time, operating modes, reset/restart rules, integrity target, diagnostics, proof testing, bypass requirements, interfaces, and environmental conditions. This list is supplemental elaboration of the lecture lifecycle, not a substitute for the standard's full SRS requirements.

Example SIF: If vessel pressure reaches the defined high-high threshold, the pressure sensing subsystem signals the safety logic solver, which commands a shutdown valve to stop feed within the required time. State whether reset is manual, whether the action latches, how faults are handled, and what happens to utilities. Without those details, “shut down on high pressure” is incomplete.

7.5 Voting and architecture—supplemental explanation

Architecture	Basic meaning	Trade-off
1oo1	One channel out of one initiates action	Simple; a dangerous channel failure can defeat detection
1oo2	Either of two channels can initiate action	Can improve dangerous-failure performance, but may increase spurious trips
2oo3	Two of three channels must vote for action	Can balance availability and fault tolerance; needs disagreement/diagnostic handling

“oo” means “out of.” Voting is not itself a SIL rating. Account for common causes, diagnostic behavior, repair, test coverage, systematic failures, and the full sensor-to-final-element chain. S3 names 1oo2 and 2oo3 for sensors and DMR/TMR for logic-solver redundancy.

7.6 Worked LOPA case

Supplemental numeric example using the lecture method: Initiating-event frequency = 0.1 per year. One justified independent non-SIS protection layer has PFD = 0.1. Frequency before adding the SIF = 0.1/year × 0.1 = 0.01/year. Tolerable frequency = 10^{-5}/year.

Required SIF PFDavg ≤ (10^{-5}/year)/(10^{-2}/year) = **10^{-3}**; required RRF ≥ **1,000**. A design at PFDavg = 10^{-3} is at the SIL 2 lower numerical boundary; a value below it is within the SIL 3 numerical band. Actual allocation must address the project acceptance rules, uncertainties, margin, and applicable process-sector constraints. Do not label a boundary by a rounded RRF chart alone.

If two purported layers share the same failing transmitter, multiplying their PFDs as if independent is not justified. LOPA estimates depend on initiating frequency, conditional modifiers where used, independence,

consequence assumptions, and valid layer credits.

7.7 Testing and maintenance distinctions

Activity	Purpose	What it does not automatically establish
Inspection	Check selection, installation, condition, documentation	Complete functional response
Calibration	Compare response to a suitable reference	Entire sensor–logic–final element action
Diagnostic testing	Detect covered faults automatically or by defined checks	Detection of every hidden dangerous failure
Functional testing	Verify specified actions and sequences	Complete numerical SIL compliance by itself
Proof testing	Reveal dangerous undetected failures according to defined coverage	Perfect detection of all possible failures
Partial-stroke test	Check part of valve movement/associated function	Full stroke, seat tightness, or all end-to-end failures
Validation	Demonstrate implemented SIF meets its safety requirements	A reason to stop lifecycle maintenance

End-to-end tests check the complete cause-to-action path. “Full proof test” still needs specified procedures, coverage, acceptance criteria, and limitations. Record as-found conditions and actual failures; an undocumented pass cannot support maintenance and performance tracking.

7.8 Why proof-test interval matters

For a simplified 1oo1 low-demand subsystem with constant dangerous-undetected failure rate λDU, perfect periodic proof testing, negligible repair time, and the relevant assumptions, **PFDavg ≈ λDU × T / 2**. This is supplemental math, not a universal full-SIF formula.

Example: $\lambda DU = 1 \times 10^{-6}$ per hour and T = 8,760 hours gives PFDavg ≈ 0.00438. Doubling T approximately doubles this contribution. Use consistent time units. For real systems include test coverage, repairs, diagnostics, architecture, common cause, other subsystems, and systematic requirements.

For small independent subsystem probabilities, a rough series approximation is PFDavg,total ≈ PFDsensor + PFDlogic + PFDfinal. For 0.001 + 0.0001 + 0.004, total ≈ 0.0051 and RRF ≈ 196. This is an illustrative calculation, not a formal verification for an actual plant.

7.9 Bypasses, changes, and records

Use authorized, time-limited bypass procedures with risk assessment, compensating measures, clear indication, responsibility, logging, and restoration checks. Never leave a test bypass active unnoticed. Before returning to service, confirm the correct configuration, enabled logic, restored connections, removed overrides, appropriate reset, and recorded results.

The lectures emphasize test dates, pass/fail results, observed field failures, and documentation. Review actual performance against assumed failure rates and test intervals. Safety-related software modifications require controlled change and relevant re-verification/revalidation; a seemingly small setpoint change can change risk.

7.10 Practice

Q37 — Remembering. Which sequence represents SIS architecture? A. Logic–final element–sensor; B. Sensor–logic solver–final element; C. Valve–PFD–ISO certificate; D. Operator–schedule–flange.

Answer: B. Detection comes before decision and physical protective action.

Q38 — Remembering. Is SIL normally assigned to the protective function or merely the PLC label?

Answer: The SIF. A component's capability is only one input to demonstrating the whole function's integrity.

Q39 — Understanding. Why does a successful valve partial-stroke test not replace every full proof test?

Answer: It exercises only part of the failure modes and movement. It may not prove full travel, tight shutoff, all accessories, sensing, logic, or the complete cause-to-action path.

Q40 — Understanding. Why must BPCS and SIS independence be examined before crediting them as two protection layers?

Answer: Shared failures can defeat both, so simply multiplying independent-layer probabilities can exaggerate risk reduction.

Q41 — Applying. A simplified 1oo1 subsystem has $\lambda DU = 2 \times 10^{-6}$/h and T = 4,000 h. Estimate its PFDavg contribution.

Answer: $\lambda DU\ T/2 = (2 \times 10^{-6})(4{,}000)/2 =$ **0.004**. This is one contribution under the stated assumptions, not proof of the complete SIF's SIL.

Q42 — Applying. A test is complete, but a force remains active in the safety PLC. Is the SIF ready for service?

Answer: No. Restore and verify the approved live configuration, remove/confirm overrides under the procedure, complete restoration tests, and document return to service.

8. Hazardous-Area Classification and Protection

Lecture basis: S5 pp. 3–40; S6 pp. 2–57. Study the classification system first, then choose equipment. An IP rating, a robust housing, or a generic “industrial” label is not proof of explosion protection.

8.1 What makes an area hazardous?

An explosive atmosphere can exist when flammable gas, vapor, mist, combustible dust, or ignitable fibers/flyings combine with air under conditions that can ignite. Classification considers release sources, material properties, ventilation, likelihood, frequency, and duration. It communicates requirements for design, installation, inspection, operation, and maintenance.

The fire triangle is fuel + oxygen + ignition source. For dust explosions, the lecture adds **dispersion** and **confinement** to form the dust-explosion pentagon. A primary explosion can disperse settled dust and trigger a much larger secondary explosion.

8.2 Zone classifications

Atmosphere occurrence	Gas/vapor zone	Combustible-dust zone
Present continuously, for long periods, or frequently	Zone 0	Zone 20
Likely occasionally during normal operation	Zone 1	Zone 21
Not likely in normal operation; short duration if it occurs	Zone 2	Zone 22

Source conflict: S5 p. 11 incorrectly labels its dust rows Zone 0/1/2. S5 p. 21 and S6 p. 12 correctly use Zone 20/21/22. Use **20/21/22 for dust**. A zone describes atmosphere occurrence, not simply the seriousness of an injury or the explosiveness of a gas.

8.3 NEC class/division and material groups

Class	Material	Common groups/examples
Class I	Flammable gases/vapors	A acetylene; B hydrogen; C ethylene; D propane—representative examples
Class II	Combustible dust	E metal dust; F carbonaceous dust; G other combustible dust such as grain
Class III	Ignitable fibers/flyings	Textile fibers, lint, similar easily ignitable materials

Division 1 generally includes hazards present in normal operation and other specified frequent/abnormal conditions. Division 2 generally covers conditions normally contained/absent but possible under specified abnormal circumstances. These are study summaries; exact definitions depend on class and adopted code.

Do not equate every detail of the gas and dust definitions.

Systems are not interchangeable: Gas Zones 0 and 1 are often broadly compared with Division 1; Zone 2 with Division 2. Dust Zone 20/21 is often broadly compared with Class II Division 1; Zone 22 with Division 2. These are **approximate teaching comparisons**, not automatic installation approvals or certification equivalence.

8.4 IEC material groups

IEC grouping	Material / representative example	Memory cue
Group I	Mining applications susceptible to firedamp	Distinct from above-ground Group II gases
IIA	Gas: propane	Less demanding gas subgroup than IIB/IIC
IIB	Gas: ethylene	Intermediate gas subgroup
IIC	Gas: hydrogen/acetylene	Most demanding of these gas subgroups for relevant protection parameters
IIIA	Combustible flyings	Dust/flyings group uses three I's
IIIB	Nonconductive dust	Grain/flour/wood examples, subject to actual material properties
IIIC	Conductive dust	Conductive dust properties determine assignment

Source conflict: S5 p. 18 incorrectly uses IIA/IIB/IIC for its dust table. S6 p. 35 correctly uses **IIIA/IIIB/IIIC**. Gas groups and dust groups must not be swapped. A Group II/III designation describes relevant material/protection characteristics; calling hydrogen universally “the most dangerous substance” oversimplifies risk.

Equipment rated IIC can generally cover IIB/IIA gas subgroup requirements where all other conditions and the certificate permit; that does **not** by itself establish correct zone/EPL, temperature, ambient range, installation, or suitability for dust. Similarly, a gas-group correspondence between IEC and NEC does not make their certifications interchangeable.

8.5 Gas properties—understanding rather than isolated numbers

Flash point is the lowest temperature at which a liquid gives off enough vapor to ignite under the specified test conditions. Autoignition temperature concerns ignition without an external spark/flame under specified conditions. LEL and UEL bound the flammable concentration range for stated conditions. Material properties and test conditions affect tabulated values.

S6 lists representative LEL/autoignition values: hydrogen 4 vol% / 560 °C, methane 4.4 vol% / 600 °C, propane 2.1 vol% / 450 °C. **Lecture-reported values, not independently validated here.** Do not use them for an actual design without an authoritative material-specific source. Hydrogen/methane tend to rise relative to air; propane tends to collect low, but ventilation, temperature, geometry, and release behavior affect actual dispersion.

Gas subgrouping relates to ignition/transmission properties such as MESG (maximum experimental safe gap) and MIC ratio (minimum igniting current ratio). The source gives simplified thresholds. Learn their purpose; do not treat a brief classroom table as a substitute for formal classification of an actual mixture.

8.6 Dust hazards

Settled dust can ignite on hot surfaces and can be re-suspended into an explosive cloud. Housekeeping and dust accumulation therefore matter even when a cloud is not continuously visible. Dust-layer thickness affects heat insulation and ignition behavior; gas T-class selection alone does not resolve dust-layer/cloud temperature requirements.

Property	Meaning	Why it matters
MEC	Minimum explosible concentration of a dust cloud under the test conditions	Whether a suspended cloud can propagate an explosion
MIE	Minimum ignition energy under the test conditions	Sensitivity to sparks/static discharge
Kst	Dust deflagration index	Relative explosion severity; informs engineered protection
Pmax	Maximum explosion pressure under test conditions	Containment/venting/suppression design input
Electrical resistivity	Conductivity-related characteristic	IEC dust subgroup selection
Particle size / moisture	Influences ignition and burning behavior	Actual process material may differ from a generic name

S6 describes Kst bands as up to 200, 201–300, and above 300. Supplemental conventional notation is **St 1: 0 < Kst ≤ 200; St 2: 200 < Kst ≤ 300; St 3: Kst > 300**, typically in bar·m/s; St 0 indicates no explosion in the specific test. Kst is not the probability of an explosion. A low value does not mean safe to ignore.

S6 discusses the 2008 Imperial Sugar event as a secondary-explosion example. The key exam lesson is the chain: primary event → settled dust dispersed → larger combustible cloud → secondary explosion. Details of accident causation require the investigation report, which was not supplied.

Additional lecture detail—S6 pp. 29, 34–35: The slides distinguish dust particles at 500 µm or less from larger flyings and use 1,000 Ω·m as the resistivity boundary: IIIB above 1,000 Ω·m, IIIC at or below it. Confirm actual material properties by appropriate data/testing rather than classifying all dust by a familiar name. The Class III slide gives 165 °C and 120 °C limits for its stated non-overload and overload-capable equipment cases; retain those as lecture-reported, code/application-dependent values rather than universal surface limits for every fiber installation.

Worked dust-temperature example from the lecture

S6 p. 31 contains a graphical calculation: cloud minimum ignition temperature = 420 °C; a 5 mm layer ignition temperature = 250 °C. The slide compares a cloud-derived limit of (2/3) × 420 = **280 °C** with a layer-derived limit of 250 − 75 = **175 °C**. The lower value, **175 °C**, governs in that example. A 150 °C marked maximum passes that comparison; 200 °C does not.

This is a specific lecture example of cloud and thin-layer assessment, not a universal formula for all dust deposits. Thicker layers, different deposition conditions, actual test data, equipment certification and the adopted standard may require additional assessment or lower limits. Do not substitute the gas T1–T6 table for the separate dust cloud/layer evaluation.

8.7 Temperature classes

Gas T-class	Maximum equipment surface temperature
T1	450 °C
T2	300 °C
T3	200 °C
T4	135 °C
T5	100 °C
T6	85 °C

Mnemonic: Higher T number → lower permitted maximum surface temperature. Equipment temperature suitability must be checked against the hazardous material's relevant ignition properties, ambient range, certificate conditions, and installation. A T4 marking is a class limit; it is not a promise that the equipment always operates at exactly 135 °C.

Example: For a gas with a stated autoignition temperature of 180 °C, T3 permits 200 °C and is unsuitable on that comparison. T4 permits 135 °C and passes that temperature comparison, but the zone/EPL, gas group, ambient and certification must also be suitable. Dust needs additional cloud/layer assessment.

8.8 Explosion-protection methods

Code	Method / principle	Key exam distinction
Ex d	Flameproof enclosure	Contain internal explosion and prevent ignition transmission to surrounding atmosphere
Ex i	Intrinsic safety	Limit ignition-capable electrical/thermal energy under specified conditions/faults
Ex e	Increased safety	Measures against ignition sources such as arcs, sparks, excessive heat within its scope
Ex p	Pressurization/purging	Protective gas keeps the hazardous atmosphere out, with required safeguards
Ex m	Encapsulation	Isolate ignition-capable components in compound
Ex o	Liquid/oil immersion	Protective liquid separates relevant ignition sources
Ex q	Powder filling	Protective filling prevents ignition transmission under specified design
Ex t	Dust protection by enclosure	Limit dust ingress and surface temperature
Ex ec / older Ex n teaching references	Enhanced/non-sparking-type gas protection context for appropriate applications	Exact designation depends on edition and certificate
Ex h	Non-electrical equipment protection framework	Mechanical equipment also presents ignition hazards

Subtypes, protection levels, application limits, and gas/dust scope matter. A method name alone does not prove suitability for every zone. Ex d is not a generic solution to dust hazards; Ex t is the dust-enclosure method in the lectures. Ex i can be applied in relevant gas/dust contexts with appropriate certification and system design.

8.9 Intrinsic safety versus flameproof protection

Intrinsic safety prevents ignition-capable energy in the circuit. The **whole circuit** matters: field device, barrier/isolator, associated apparatus, cable capacitance/inductance, grounding where required, segregation, and installation conditions. A low voltage alone does not prove intrinsic safety.

Flameproof protection permits an internal ignition while preventing it from igniting the external atmosphere through the designed enclosure/flamepaths. Preserve enclosure integrity, correct entries/glands, fasteners, seals where required, and installation instructions. Do not drill, modify, or substitute parts without the necessary approved assessment.

Terminology qualification: IEC “flameproof Ex d” and NEC “explosionproof” share a containment concept, but their standards, tests, markings, and approvals are not automatically interchangeable.

8.10 Reading a marking and selecting equipment

The lecture example **Ex d IIC T4 Gb** indicates Ex protection, flameproof method, IIC gas subgroup, T4 temperature class, and gas EPL Gb. Under typical zone/EPL allocation, Ga is associated with Zone 0, Gb with Zone 1, and Gc with Zone 2; higher protection can cover less demanding zones where suitable. Dust EPLs Da/Db/Dc correspond in that typical allocation to Zones 20/21/22.

Do not overread the example: “Ex” is a marking element, not proof of approval by itself. Gb does not make equipment suitable for Zone 0. Check the actual certificate, protection subtype, ambient range, special conditions, material group, temperature, installation accessories, and local adoption. Modern certificate markings may use more detailed subtype notation than the lecture's simplified “d.”

Selection sequence: identify the classified area → identify the actual material and group → verify temperature requirements → choose protection method and required EPL/approval → check certification and special conditions → verify the complete installation. Location classification must precede equipment shopping.

8.11 Practice

Q43 — Remembering. Which dust zone has a cloud present continuously, frequently, or for long periods?

Answer: Zone 20. Zone 0 is the corresponding gas/vapor occurrence category, not the dust label.

Q44 — Remembering. What is the maximum equipment surface temperature for T4?

Answer: 135 °C. T3 is 200 °C; T5 is 100 °C.

Q45 — Understanding. Why is an IP66 enclosure not automatically acceptable in Zone 1?

Answer: IP describes ingress protection. Explosion protection additionally requires the appropriate protection concept, EPL/area suitability, gas group, temperature/ambient limits, certification, and installation.

Q46 — Understanding. What is the central difference between Ex i and Ex d?

Answer: Ex i limits ignition-capable energy in the circuit; Ex d contains an internal explosion and prevents ignition transmission. Neither is established by appearance alone.

Q47 — Applying. Can a device marked Ex d IIC T4 Gb be selected for Zone 0 solely because it has the most demanding gas subgroup?

Answer: No. IIC addresses gas subgroup, while Gb is normally allocated to Zone 1—not Zone 0. These are separate selection dimensions.

Q48 — Applying. A dust installation specification says “Zone 1, IIB” for nonconductive grain dust. What needs clarification?

Answer: It mixes gas labels with a dust application. Determine the correct dust zone (20/21/22), dust subgroup (often IIIB for nonconductive dust, subject to material properties), temperature requirements, and certified protection. Do not just replace labels without classification assessment.

9. Electrical and Instrumentation Installation Requirements

Lecture basis: S5 pp. 41–79; S6 pp. 59–91. Installation protects people, equipment, reliable operation, signal integrity, maintainability, and compliance. Correct equipment selection can be defeated by incorrect installation.

9.1 Electrical wiring versus instrumentation wiring

Aspect	Electrical power wiring	Instrumentation wiring
Main purpose	Deliver and control power	Carry measurement/control information and device power where applicable
Typical devices	Motors, heaters, panels, feeders	Transmitters, switches, RTDs/thermocouples, PLC/DCS I/O, control valves
Typical concerns	Ampacity, short circuit, voltage drop, insulation, protective devices	Signal type, polarity, range, shielding, segregation, continuity, noise
Path example	Source → protection → control → load	Instrument → JB → terminals → PLC/DCS

The lectures mention 120/240 V AC and higher distribution voltages as examples; they are not a universal plant-voltage requirement. Instrument wiring still needs electrical safety and suitable current/voltage ratings, and some instrument circuits carry both power and signal.

9.2 Cable selection and segregation

Select conductor size/material for current and voltage drop; insulation/jacket for voltage, temperature, chemicals, fire/environment requirements; shielding for relevant interference; and armor for required mechanical protection. Verify compatibility with the signal technology and manufacturer instructions. An RTD, thermocouple, Ethernet cable, and 4–20 mA circuit do not have identical requirements.

The lecture grouping is power/high voltage, AC/DC control, analog/low-level signals, and data/communications. Separate incompatible circuits using the appropriate routes, trays, raceways, barriers, or spacing. Reduce long parallel exposure to strong interference sources; crossing at approximately 90° can reduce coupling where crossing is necessary.

Flag: The source says cables must cross “exclusively” at 90° and gives 300 mm separation. These are simplified practices, not universally applicable numerical code requirements. Required separation depends on voltage, shielding, raceway, circuit classification, EMC design, manufacturer, and adopted standards.

9.3 Grounding and bonding

Concept	Primary meaning	What to remember
Grounding/earthing	Connection to earth or the relevant grounding system as defined	Reference, surge/static behavior, and electrical-safety role depend on system design
Bonding	Conductive interconnection of metal parts	Maintain continuity and limit dangerous potential differences
Protective earth—PE	Protective arrangement for exposed conductive parts	Must not be defeated to fix noise
Signal/reference grounding	Controlled reference for sensitive electronics	Coordinate with safety bonding and equipment instructions
Shield termination	Connection arrangement for cable shield	Depends on signal frequency, equipment and EMC/IS design

Important clarification: A separate earth rod alone is not generally an adequate low-impedance metallic return path for clearing a line-to-case fault in typical systems. Protective bonding/conductors must provide the required return path to the source/protection arrangement. Do not interpret the lecture's "earth = 0 V" wording as a guarantee that all ground points are always at identical potential.

"Clean ground" does not mean leave exposed metal unbonded or create an uncontrolled isolated earth. A signal reference can be arranged to reduce noise while preserving the required protective equipotential bonding. Safety PE and signal/shield functions are related but not interchangeable.

9.4 Shielding and noise reduction

Technique	Main benefit	Limitation/caution
Foil shield	High coverage, often useful for high-frequency interference	Termination and mechanical construction matter
Braided shield	Durable, flexible conductive mesh	Coverage and frequency response differ from foil
Foil + braid	Combines coverage with mechanical/electrical benefits	More construction complexity; still needs correct termination
Twisted pair	Reduces loop area and pickup from magnetic coupling	Must maintain pair geometry and suitable terminations
Separation/routing	Reduce coupling from noise sources	Distances must follow actual design requirements
Filtering and grounding/bonding	Control conducted noise and reference differences	Wrong arrangement can create other paths/problems

Source qualification: S5 p. 57 recommends grounding shields at one end, usually the panel. This is a common low-frequency analog technique for avoiding certain ground-loop currents, **not a universal rule**. High-frequency communications/EMC designs may require both-end or multipoint bonding and proper high-frequency terminations. Follow equipment, system, intrinsic-safety, and EMC requirements. Never disconnect PE as a troubleshooting shortcut.

9.5 Conduit, cable tray, and cable gland

Item	Purpose	Lecture examples
Conduit	Enclosed pathway for routing and protecting wires/cables	RMC, EMT, PVC/HDPE, flexible conduit
Cable tray/ladder	Support and organize cable runs	Ladder, perforated, solid bottom, channel
Cable gland	Secure/seal cable entry to an enclosure; provide required retention/bonding features	General-purpose, armored, suitable Ex entries

A gland is **not an enclosed raceway**; that wording in S6 p. 79 duplicates the conduit definition. Match cable type, diameter, armor, thread, enclosure, IP/environment, hazardous-area protection, certificate, and installation instructions. An intrinsically safe circuit still needs appropriate entry, segregation, and enclosure arrangements; a generic “Ex i gland” label is not a substitute for assessment.

Support trays and conduits for their loads, protect against corrosion/mechanical damage, preserve bend radius, account for expansion/vibration, bond metalwork as required, and fire-stop rated penetrations with an approved suitable system. A tray may serve as an equipment grounding conductor only when all applicable conditions are met; do not assume every metal tray qualifies.

9.6 Source numbers that must be treated conditionally

Lecture assertion—S6	Why it must be qualified
Conduit fill ≤40% power / ≤30% instrument	Not a universal split by cable purpose; permitted fill depends on applicable raceway/cable rules and conditions
Bend radius ≥6× diameter; shielded ≥10×	Product construction, installation/operating condition, and manufacturer requirements control
Signal/power separation ≥300 mm	Depends on circuits, barriers, raceway, shielding, and EMC requirements
Supports within 1 m of fittings; RMC ≤3 m	Raceway type, code edition, exceptions, and installation conditions matter
Tray access clearance ≥300 mm	Project/access/code-specific; not a universal rule for every tray
Panel access ≥1 m front / ≥0.8 m rear	Depends on voltage, exposure, layout, service, and adopted requirements
Indoor IP30–IP54 / outdoor IP65–IP66	Selection examples, not blanket requirements or proof of Ex suitability

Keep these as **lecture-reported values for instructor clarification**, not universally verified code limits. They remain preserved in the source appendix. No unverified numerical value should be used as an actual construction criterion.

9.7 Panel installation

A panel may house PLCs, power supplies, I/O, relays, terminals, instruments, and protective devices. Lecture types include main control panel, local control panel, junction/instrument box, and HMI/operator station. Their roles overlap but an HMI alone is not the whole control system.

Verify environment/IP rating, hazardous-area suitability, structural support, access, heat dissipation, corrosion/vibration exposure, routing/segregation, wire identification, terminal arrangement, protective

bonding, and maintenance access. Electrical and pneumatic/mechanical installation must agree with the hook-up, layout, wiring, and loop documents.

The lecture's five-stage installation lifecycle is planning/design → procurement/delivery → physical/electrical installation → commissioning/calibration → maintenance/monitoring. This installation summary is distinct from the full functional-safety lifecycle.

9.8 Inspection, testing, commissioning

Inspection checks that selection, installation, condition, identification and workmanship match approved requirements. Testing demonstrates relevant electrical/signal/function behavior under a defined procedure. Commissioning prepares and verifies the integrated system for actual operation. The stages may include repeated checks as defects are corrected; headings alone do not establish completion.

Check	What it establishes	Caution
Visual/document inspection	Correct equipment, labels, routing, connections, certificates, drawings	Does not prove measurement accuracy
Continuity/polarity	Intended conductive path and orientation	A shorted wrong circuit can still show continuity
Insulation testing	Insulation integrity under the specified method	Isolate sensitive electronics/IS apparatus as required; do not indiscriminately megger them
Ground/bond continuity	Required conductive protective interconnections	Reference limits/procedure must be applicable
Calibration	Accuracy/response against reference and acceptance criteria	Distinct from adjustment and end-to-end function
Loop check	Correct path, tag, I/O, scaling, display, output and response as included	Define scope; not automatically complete SIF proof testing
Interlock/trip/cause-and-effect test	Required initiating condition causes the specified action	Use approved safe conditions and restore overrides
Commissioning/handover	Integrated readiness with records and resolved deficiencies	Outstanding critical defects prevent release

Source inconsistency: S5 p. 78 has a testing heading but begins with a commissioning definition and ends with “before commissioning.” This reviewer distinguishes the three activities rather than reproducing that wording as a coherent definition.

9.9 Standards and legal-reference corrections

Reference	Study purpose / correction
IEC 60079-0	General requirements for Ex equipment
IEC 60079-10-1 / -10-2	Gas/vapor and dust area classification, respectively
IEC 60079-14	Electrical installation design, selection, erection and initial inspection context; title varies by edition
IEC 60079-17	Inspection and maintenance of Ex electrical installations
IEC 61326-1	EMC for measurement, control and laboratory equipment
IEC TR 61000-5-2	Earthing and cabling guidance for EMC
IEC 61386	Conduit systems
IEC 61537	Cable tray/ladder systems
NEMA VE 1	Metal cable tray systems; distinguish construction standard from installation guidance
IEC 60529	IP code / degrees of protection by enclosures; not the machinery electrical-equipment standard
IEC 62444	Cable-gland requirements—supplemental correction to S6's IEC 61010-1 gland attribution
IEC 61010-1	Safety of measurement, control, and laboratory electrical equipment within scope; not a universal gland standard
IEC 60204-1	Electrical equipment of machines—supplemental identification where relevant
IEC 62208	Empty enclosures for low-voltage switchgear/controlgear assemblies within scope
ISA 5.1	Instrumentation symbols/identification; not a universal wire-labeling code
ISA RP12.06.01	Intrinsically safe system wiring/installation practices; not all general wiring practice
PEC / NEC (NFPA 70)	Apply the locally adopted requirements and actual scope; do not treat every foreign code as automatically local law
Republic Act 9292	Electronics Engineering Law of 2004; S6 incorrectly calls it the Electrical Engineering Law
Republic Act 7920	New Electrical Engineering Law—supplemental correction; legal scope/licensing needs authoritative review

S6 p. 74 associates NEC Articles 342/344/350 with RMC/EMT/flexible conduit. As a study correction, 342 concerns intermediate metal conduit, 344 rigid metal conduit, 350 liquidtight flexible metal conduit, and 358 EMT in commonly used NEC editions. Verify the adopted edition and actual application. S6's description of PEC Part 2 as simply “industrial installations” is oversimplified; do not use that wording to determine legal scope.

These reference corrections are editorial technical background, not a claim that every referenced edition/title/legal requirement was independently checked online in this session. For actual compliance, consult the adopted documents and competent responsible professionals.

9.10 Practice

Q49 — Remembering. Distinguish grounding and bonding.

Answer: Grounding connects to earth/the relevant grounding system; bonding interconnects conductive parts to maintain continuity and limit potential differences. Correct protective arrangements depend on system design.

Q50 — Remembering. Which standard identifies enclosure ingress protection? A. IEC 60529; B. ISO 9001; C. ASME B31.3; D. ISA 5.2.

Answer: A. IEC 60529 addresses IP classification. It does not by itself establish explosion protection.

Q51 — Understanding. Why should a cable shield not be used as a casual substitute for a protective conductor?

Answer: Signal shielding and protective fault-current duties have different requirements. A shield may not have the capacity, continuity, connections, or approval needed for protective bonding.

Q52 — Understanding. Is grounding every shield at one end always correct?

Answer: No. It is a common technique in some low-frequency analog systems. High-frequency, network, EMC, equipment, and intrinsic-safety requirements can call for other arrangements.

Q53 — Applying. A transmitter is accurate locally but its PLC reading jumps whenever a motor starts. What should be investigated?

Answer: Noise coupling, routing/segregation, shield termination, bonding/reference differences, supply quality, input configuration and wiring. Compare signal measurements safely; do not disconnect PE to make the reading steady.

Q54 — Applying. A general-purpose gland fits the cable diameter and maintains IP66. Is it automatically acceptable for an Ex d enclosure?

Answer: No. Check Ex certification, protection compatibility, thread, cable construction, required sealing/entry arrangement, enclosure conditions and manufacturer instructions. Fit and IP rating alone are insufficient.

10. Integrated Application Cases

These newly written cases combine the exam's remembering, understanding, and applying levels. State the principle, identify the relevant document/requirement, apply it to the facts, and explain any limitation.

Case 1—ASME / instrument pressure boundary

Q55. A purchase request specifies only “PT-204, 0–20 bar, 4–20 mA.” The fluid is corrosive and the line is hot. What is missing?

Model answer and rationale: Confirm process/design pressure and temperature, fluid compatibility, wetted materials, connection rating/type, manifold/tubing/seal requirements, accuracy, electrical supply, environmental/Ex suitability, and the governing piping specification/code. Range and output alone do not define safe pressure-boundary selection. Cross-check the data sheet, P&ID, hook-up and piping specification. This ASME-oriented case is supplemental because no ASME lecture was supplied.

Case 2—ISO / recurrent calibration errors

Q56. Three transmitters are adjusted after failing calibration. A supervisor closes the issue because all now pass. What quality action remains?

Model answer and rationale: Preserve as-found/as-left records, assess prior measurement impact, investigate common causes, determine whether corrective action is needed, implement suitable changes, and verify effectiveness. Adjustment is a correction; it does not establish why three failures occurred or whether the process will recur. This ISO-oriented case is supplemental.

Case 3—ISA / interpreting a loop

Q57. A diagram shows FT-210 → FIC-210 → FV-210, with a dashed electrical signal style defined in its legend and FC on the valve. Explain the loop and the limits of the drawing.

Model answer and rationale: FT transmits flow; FIC indicates/controls flow; FV changes flow in loop 210. The defined line style conveys electrical information, not the process-fluid route. FC specifies closure on the identified failure. Check exact wiring in the loop diagram and actual actuator/accessory behavior in technical documents; do not infer terminals or all failure scenarios from the P&ID alone.

Case 4—control / scaling and response

Q58. A linear level transmitter is configured 1–5 m. At 12 mA the PLC shows 2.5 m. The inlet valve is stable. What should you check before retuning the controller?

Model answer and rationale: Expected level is $1 + 0.5 \times (5 - 1) = \mathbf{3\ m}$. Check input range, engineering-unit scaling, LRV/span, transmitter configuration and signal measurement. The wrong display suggests scaling/measurement issues; tuning cannot correct a wrong PV conversion.

Case 5—documentation / conflicting revisions

Q59. A technician has loop drawing Rev 2, the PLC I/O list is Rev 4, and the P&ID recently added a transmitter. How should work proceed?

Model answer and rationale: Confirm the current approved controlled document set and affected change package. Resolve tag, channel, terminal, range and cable discrepancies; obtain checked/approved updates; distribute the correct issue and retain traceability. Different revision numbers are not automatically wrong because documents evolve independently, but inconsistent content must be resolved before relying on it.

Case 6—functional safety / incomplete integrity claim

Q60. A supplier says “this is a SIL 3 sensor, so your shutdown loop is SIL 3.” Evaluate.

Model answer and rationale: A sensor's assessed capability does not establish the entire function's integrity. Define the SIF/target and evaluate sensors, logic, final elements, utilities, architecture, common causes, application limits, proof testing, failure rates, systematic capability, and lifecycle requirements. The claim confuses component capability with complete SIF achievement.

Case 7—SIS / proof-test change

Q61. Operations wants to double a proof-test interval because no failures were recorded last year. What assessment is needed?

Model answer and rationale: Review the reliability calculation and the validity/completeness of field data, coverage, diagnostics, repairs, and all affected assumptions. Under the simple λDU T/2 model, doubling the interval doubles that dangerous-undetected contribution. Apply management of change, authorize the revised procedure only after the target integrity remains justified, and update records. One year without observed failure is not proof that no hidden failures exist.

Case 8—hazardous area / separate selection dimensions

Q62. A device is Ex d IIC T3 Gb. The area is Zone 1 with a gas autoignition temperature stated as 180 °C. Is the device suitable?

Model answer and rationale: Gb is normally consistent with Zone 1 and IIC may satisfy the relevant subgroup, subject to certificate conditions. But T3 permits a surface temperature of 200 °C, which fails the stated 180 °C comparison. Select a suitable lower surface-temperature class, such as T4 on this comparison, and still check ambient range, protection/certification, installation and special conditions. One suitable marking element cannot compensate for another unsuitable one.

Case 9—installation / noise versus safety

Q63. A technician removes the panel PE connection and the noisy signal becomes stable. Can the panel be released?

Model answer and rationale: No. Restore the required protective connection and diagnose routing, shields, reference paths, supply quality and equipment configuration using approved methods. Signal stability does not justify loss of shock protection. Reinspect, retest, document the resolved cause, and verify required protection before release.

11. Answers and Rationales for the Uploaded 20-Item Quiz

Source: S4 pp. 32–37. The original wording/options are preserved in the Source Appendix. The following key is worked out from the question content; the uploads do not provide a separate official answer key. Items Q1–Q20 in this section refer to the source quiz, not reviewer Q1–Q63.

Source item	Answer	Rationale
1	B	Functional safety provides the required response to hazardous conditions and reduces risk; production optimization is not its primary objective.
2	C—IEC 61508	This is the general E/E/PE functional-safety standard. The other choices are altered numbers.
3	C—Safety Integrity Level	“Safely,” “System Integrity Logic,” and “Safety Installation Level” are incorrect expansions.
4	D—SIL 4	Highest of the four IEC 61508 integrity levels; do not confuse this with ordinary IEC 61511 process-SIF targeting.
5	B	Risk assessment evaluates hazards and the risk reduction still needed. Routine production performance alone is not enough.
6	A	Safety requirements are derived from hazard/risk analysis as part of the lifecycle. The other choices focus on ordinary operating performance.
7	B	An SIS provides a protective layer against hazardous events when appropriately designed; it does not replace every BPCS function.
8	B	The basic sequence is sensor → logic solver → final element.
9	A	A SIF detects a specified hazard and performs a defined protective action toward a predetermined safe state.
10	B	Proof testing checks the ability to perform the required safety function and identifies covered hidden failures; it does not eliminate maintenance.
11	SIS—true	Functional safety reduces risk through correct safety-related response. “Acceptable/tolerable” is context-dependent wording.

Source item	Answer	Rationale
12	SIS—true	IEC 61508 provides the general E/E/PE functional-safety framework.
13	SIS—true	A higher SIL generally requires greater integrity/risk reduction. It does not simply state consequence severity.
14	SIL—false	Risk assessment is necessary to identify hazards and determine required measures/integrity.
15	SIS—true	Safety management extends from early hazard analysis through operation, maintenance and decommissioning.
16	SIL—false	IEC 61511 focuses on process-sector SIS, not general-purpose production-control systems.
17	SIL—false	The statement says “an SIL consists of sensors, logic solver and final elements.” That describes an SIS/SIF architecture; SIL is an integrity level.
18	SIS—true	The logic solver receives inputs, executes defined safety logic, and commands final elements.
19	SIS—true	Proof tests are intended to reveal covered dangerous failures missed by normal operation/diagnostics.
20	SIS—true	Regular suitable testing and maintenance support the SIF's required integrity over its life.

Exam trap: In this particular source quiz, “SIS” means true and “SIL” means false because the instructions explicitly assign those answer tokens. That is only a response convention; it does not redefine the technical terms.

12. Source Conflicts, Unsupported Claims, and Study Limits

Use this register when a source slide and the reviewer differ. Original source statements are preserved; the distinction between correction, qualification, and missing support is deliberate.

Issue and location	Treatment in this reviewer
ASME / ISO coverage absent from S1–S6	Chapters 1–2 are supplemental. Instructor-specific scope and code editions remain unconfirmed.
Mixed ISA editions—S2 slides 2, 10, 47	Use the adopted edition/legend; no single edition is silently imposed on every figure.
“Three-letter” tag model—S2 graphics	Teaching simplification; tags vary in length and modifier position.
Safety-loop numbers 900–999—S2 slide 21	Possible project convention, not a universal mandatory range.
Every FO/FC action attributed to a spring—S2 slide 18	Actual actuator/energy/accessory design and specified failure control the behavior.
“Safety Integrated Level”—S3 p. 28	Corrected to Safety Integrity Level.
SRS used for safety-related system—S4 p. 5	Context distinguished from Safety Requirements Specification.
SIL 4 upper RRF printed “100,00”—S4 p. 6	Apparent typo; use 100,000 in the low-demand numerical explanation.
PFD alone described as proving hardware/software reliability—S4 p. 7	PFDavg is one metric; software/systematic integrity requires more than probability calculation.
LOPA called exact—S3 p. 47; S4 p. 7	Semi-quantitative, assumption-dependent estimate; do not claim exact prediction.
“At least 100 times” automatically labeled SIL 2—S3 p. 43	Boundaries clarified using PFDavg inequalities.
Blanket physical separation of SIS/BPCS—S3 pp. 22, 44	Independence/common-cause requirements must be assessed under the adopted standard.
Final element always highest PFD—S3 p. 39	Common practical concern, not universal; verify actual contributions.
Dust rows named Zone 0/1/2—S5 p. 11	Correct dust labels are 20/21/22; supported by S5 p. 21 and S6 p. 12.
Dust groups named IIA/IIB/IIC—S5 p. 18	Correct dust labels IIIA/IIIB/IIIC; supported by S6 p. 35.
“Most dangerous” gas ranking—S6 pp. 21–23, 56	Reframed as protection subgroup demand; overall risk has additional dimensions.
IEC/NEC equivalence—S6 pp. 23, 36, 50	Approximate conceptual mapping; not automatic approval/certification equivalence.

Issue and location	Treatment in this reviewer
Temperature classes applied generically to dust—S5 p. 23	Gas T-class table retained; dust cloud/layer temperature assessment distinguished.
“Ex” automatically means approved—S6 p. 57	Check actual certification/conditions; marking fragment alone is insufficient.
Ground always 0 V / earth alone clears faults—S5 pp. 54–57	Reference potentials and protective return-path requirements clarified.
Shields always grounded one end—S5 p. 57	Conditional low-frequency analog practice; EMC/network/IS needs may differ.
Numeric fill/radius/spacing/access values—S6 pp. 75, 78, 85	Preserved as lecture claims; not verified universal requirements.
Gland described as enclosed raceway—S6 p. 79	Corrected to cable-entry/retention/sealing interface.
IEC 61010-1 called gland standard—S6 p. 80	Flagged; IEC 62444 identified as supplemental gland reference.
IEC 60529 called machinery electrical-equipment standard—S6 p. 82	Corrected to IP code; IEC 60204-1 identified where relevant.
ISA 5.1 called general wire-labeling standard—S6 p. 83	Scope limited to instrumentation symbols/identification.
RA 9292 called Electrical Engineering Law—S6 p. 83	Corrected to Electronics Engineering Law; RA 7920 identified separately.
NEC conduit article mapping—S6 p. 74	Common article identities clarified; adopted edition still controls.
Testing heading with commissioning definition—S5 p. 78	Inspection, testing and commissioning separately explained.
Exact gas-property/particle/temperature rules	Source figures/data retained; no independent material-specific design verification claimed.
Handwritten question ranges overlap at 81	No invented official question count or topic weighting.

What remains unconfirmed: The instructor's ASME/ISO lesson content; exact adopted code editions; whether the handwritten ranges contain a typo; source numerical installation rules and gas-property values as design requirements; and any official key beyond the quiz content itself. These gaps do not prevent studying the supported concepts, but they should not be hidden.

13. Final Quick Reviewer

Use this section after studying the full chapters. Cover the right-hand column and try to answer from memory.

13.1 The nine-topic memory sheet

Topic	Essential recall
ASME / piping—supplemental	B31.3 process piping; B31.1 power piping; B16 component standards; BPVC VIII vessels; IX qualifications. Verify service, materials, pressure–temperature ratings, fabrication, examination and testing.
ISO / QMS—supplemental	ISO 9001 requirements; ISO 9000 vocabulary; ISO 19011 audits; ISO/IEC 17025 lab competence. PDCA. Correction fixes the problem; corrective action addresses its cause.
ISA 5.1	First letter = variable; succeeding letters = functions/modifiers; number = loop. TIC = temperature indicating controller. Same letter can mean different things in different positions.
Process control	PV measured; SP target; MV manipulated; disturbance external. Measurement → controller → final element → process. Check mode/scaling/action before tuning.
Documentation	PFD overall process; P&ID functional connections; index inventory; data sheet specification; loop circuit; hook-up mounting; narrative behavior; cause-and-effect actions; revision control current truth.
IEC 61508	General E/E/PE functional safety. Risk reduction, safety lifecycle, random hardware and systematic failures. SIL belongs to a function; probability alone is insufficient.
IEC 61511	Process-sector SIS. Sensor → logic solver → final element. SIF is one protective function. Independence, proof tests, bypass control and management of change matter.
Hazardous areas	Gas 0/1/2; dust 20/21/22. Gas IIA/IIB/IIC; dust IIIA/IIIB/IIIC. NEC I gas, II dust, III fibers. Match location, group, temperature, protection, certification and installation.
Installation	Select/rate/route cables; segregate; bond/ground; shield correctly; use suitable glands/enclosures; inspect, test, commission, document. IP is not Ex.

13.2 Must-remember numbers and equations

Item	Quick reference
T1 / T2 / T3	450 / 300 / 200 °C maximum surface temperature
T4 / T5 / T6	135 / 100 / 85 °C maximum surface temperature
4–20 mA fraction	(I − 4)/16 for a linear configured transfer
Engineering value	LRV + fraction × span
Span	URV − LRV
Low-demand RRF	Approximately 1/PFDavg
SIL 1 PFDavg	$10^{-2} \leq PFDavg < 10^{-1}$
SIL 2 PFDavg	$10^{-3} \leq PFDavg < 10^{-2}$
SIL 3 PFDavg	$10^{-4} \leq PFDavg < 10^{-3}$
SIL 4 PFDavg—IEC 61508	$10^{-5} \leq PFDavg < 10^{-4}$
Simplified 1oo1 contribution	PFDavg ≈ λDU T/2 under its stated assumptions
Simplified LOPA	Event frequency ≈ initiating frequency × valid independent-layer PFDs

13.3 Fast document selector

“What instruments exist?” → **Index**. “Which materials/range/accuracy?” → **Data sheet**. “How does the process connect?” → **P&ID**. “Which exact wires/terminals?” → **Loop/wiring/termination drawing**. “How is it mounted?” → **Hook-up**. “Which channel?” → **I/O list**. “What happens when X occurs?” → **Narrative/logic/cause-and-effect**. “What was actually tested?” → **Calibration/test records**.

13.4 High-value contrasts

SIS = protective system; SIF = specific protective function; SIL = required integrity. BPCS = ordinary process control. IEC 61508 = general framework; IEC 61511 = process-sector SIS. Calibration = measurement comparison; adjustment = change response; loop check = signal path; proof test = uncover covered hidden dangerous failures. Grounding = connection to earth/system; bonding = connect metal parts; shielding = reduce interference. Ex i = energy limitation; Ex d = containment; Ex p = protective pressure; Ex t = dust enclosure.

13.5 Last-minute traps

- A correct range and output do not establish process-material or pressure-boundary suitability.
- A pressure class is not a fixed psi limit for every material/temperature.
- ISO 9001 certification does not certify every product's accuracy or Ex suitability.
- Tags are not always exactly three letters; first T and succeeding T differ.
- A P&ID is not a complete terminal wiring drawing.
- A safety PLC certificate does not prove the whole SIF's SIL.

- A higher SIL is not simply “more severe consequence.”
- RRF/PFD boundary inequalities matter; rounded charts can mislead.
- More redundancy does not remove common-cause or systematic failures.
- Partial-stroke testing does not prove every full-shutdown failure mode.
- Gas Zone 0 is not dust Zone 20; gas IIC is not dust IIIC.
- IP rating does not equal explosion protection.
- IIC does not make a Gb device suitable for Zone 0.
- T6 permits a cooler surface than T1.
- One-end shield grounding is conditional; never lift PE to fix noise.
- Lecture clearances/fill rules are not universal code limits without context.
- Return-to-service includes restoring bypasses/forces and documentation.

13.6 A compact answer method for application questions

Identify the hazard or required outcome. Name the relevant principle/standard and document. Apply it to the actual facts, including calculations and units. Check the whole chain and independent requirements. State any assumption or unsupported information. For example, a correct gas group does not override a wrong temperature class; a calibrated sensor does not prove a working trip; and a matching tag does not prove correct wiring.

13.7 Final self-test

Without looking: decode FIC-101 and PDT-101; list the three SIS elements; distinguish SIS/SIF/SIL; name the dust zones and groups; recite T1–T6; calculate 12 mA on a −50 to 150 °C range; identify the document for a terminal connection; explain proof-test interval impact; distinguish correction from corrective action; and name the process-piping code normally considered for a chemical-process line. If any answer is uncertain, return to that chapter before rereading the whole guide.

Self-test key: Flow indicating controller; differential-pressure transmitter; sensor/logic solver/final element; system/function/integrity level; 20/21/22 and IIIA/IIIB/IIIC; 450/300/200/135/100/85 °C; 50 °C; loop/termination drawing; longer interval generally raises the hidden-dangerous-failure contribution; fix observed issue versus address cause; ASME B31.3, subject to actual scope/adoption.